泰州学院

德国 ASIIN 工程认证专家进校考察

计算机科学与技术专业

学生座谈会问答材料

2026

学生问题

1. 专业定位与目标

问题 1：你认为本专业重点培养了哪些核心能力？

答案：
本专业主要培养我们成为具备扎实理论基础、工程实践能力和跨领域协作能力的应用型计算机工程技术人才。专业的学习成果，也就是 PLOs，包括 11 个方面：工程知识、问题分析、设计/开发解决方案、研究、使用现代工具、工程与可持续发展、工程伦理和职业规范、个人与团队、沟通、项目管理和终身学习。

从学生角度来看，这些能力不是只停留在文件里，而是通过课程和实践逐步培养出来的。例如，在《C 程序设计》中，我们需要独立完成小型系统开发，从功能设计、代码实现到调试测试都要自己完成；在《数据结构》和《算法设计与分析》中，我们训练抽象问题、设计算法和分析复杂度的能力；在《计算机网络》中，我们通过实验或项目理解网络配置、通信协议和系统运行过程；在《软件工程》和《智能应用系统开发》中，我们进一步学习需求分析、系统设计、团队协作和项目管理。因此，我认为本专业重点培养的是理论基础、工程实践、系统设计、问题解决、团队协作、沟通表达和持续学习能力。

问题 2：你如何获取培养方案、培养目标、毕业要求和课程体系？

答案：
我们主要通过几个渠道获取这些信息。第一，可以通过学校教学管理系统查看培养方案、课程安排和修读要求；第二，每门课程的教学大纲中都会说明课程目标、课程内容、考核方式以及该课程支撑哪些毕业学习成果；第三，在课程开始时，任课教师通常会介绍课程目标、学习要求、考核比例和课程与专业能力培养之间的关系；第四，学院也会通过班会、专业介绍、导师指导等方式帮助我们理解专业培养目标和课程体系。

因此，我们可以比较清楚地知道本专业要培养什么能力、需要学习哪些课程，以及每门课程在能力培养中的作用。

问题 3：入学前后你对本专业的理解有哪些变化？

答案：
入学前，我对计算机科学与技术专业的理解比较简单，主要认为它就是学习编程、写代码。进入大学后，经过课程学习和实践训练，我逐渐认识到这个专业不仅仅

Student Issue

1. Professional Focus and Objectives

Question 1: Which Core Competencies Do You Believe This Program Focuses on Developing?

answer：

This program is designed to cultivate applied computer engineering professionals with a solid theoretical foundation, strong practical engineering skills, and cross-disciplinary collaboration capabilities. The program's learning outcomes, known as PLOs, encompass 11 key areas: engineering knowledge, problem analysis, solution design/development, research, use of modern tools, engineering and sustainable development, engineering ethics and professional standards, individual and team performance, communication, project management, and lifelong learning.

From the student's perspective, these competencies are not merely theoretical concepts but are progressively developed through coursework and practical experience. For instance, in *C Programming*, students independently complete small-scale system development, handling everything from functional design and code implementation to debugging and testing. In *Data Structures* and *Algorithm Design and Analysis*, we cultivate the ability to abstract problems, design algorithms, and analyze computational complexity. Through experiments or projects in *Computer Networks*, we gain insights into network configurations, communication protocols, and system operation. In *Software Engineering* and *Intelligent Application System Development*, we further master requirements analysis, system design, teamwork, and project management. Therefore, I believe this program focuses on cultivating a solid theoretical foundation, engineering practice skills, system design capabilities, problem-solving abilities, teamwork proficiency, communication skills, and continuous learning capacity.

Question 2: How Can I Obtain the Training Program, Training Objectives, Graduation Requirements, and Curriculum System?

answer：

We primarily obtain this information through several channels. First, the training program, course schedule, and enrollment requirements can be accessed via the university's teaching management system. Second, each course syllabus outlines the course objectives, content, assessment methods, and the specific graduate learning outcomes the course supports. Third, at the beginning of the course, the instructor typically explains the course objectives, learning requirements, assessment criteria, and the course's relevance to professional competency development. Fourth, the college also assists us in understanding the program's educational goals and curriculum structure through class meetings, program introductions, and mentor guidance.

Therefore, we can clearly understand what competencies this program aims to cultivate, which courses are required, and the role of each course in fostering these competencies.

Question 3: How Has Your Understanding of Your Major Changed Before and After Enrollment?

answer：

Before enrolling, my understanding of the Computer Science and Technology major was quite simplistic—I primarily viewed it as merely learning programming and writing code. After entering university and undergoing coursework and practical training, I gradually realized that this major encompasses far more than just that.

是编程，还包括数学基础、计算机系统、算法设计、数据库、网络、操作系统、软件工程、人工智能等多个方面。

特别是通过项目化教学和课程设计，我对专业的理解发生了明显变化。例如，在《C 程序设计》课程项目中，我们不仅要写程序，还要考虑功能设计、模块划分、输入输出、异常处理和调试；在《计算机网络》课程中，我们通过实验理解网络协议和网络系统配置；在《软件工程》课程中，我们学习需求分析、系统设计、测试和文档编写。这些经历让我认识到，计算机专业更强调系统思维、工程实践和解决复杂问题的能力。

问题 4：你认为本专业毕业后可以从事哪些工作？

答案：
本专业培养的是应用型计算机工程技术人才。毕业后可以在 IT 行业、科研机构、企事业单位和公共服务部门从事计算机应用相关工作，例如软件开发、系统设计、Web 开发、移动应用开发、数据库管理、算法设计、数据处理与分析、人工智能应用开发、系统测试、网络管理和项目管理等岗位。

随着工作经验积累，毕业生也可以成长为技术骨干、系统架构人员或项目管理人员。

问题 5：你如何理解工程师的责任？

答案：
我认为工程师不仅要掌握技术，更要有社会责任感和工程伦理意识。对于计算机专业来说，我们设计和开发的软件系统、数据系统、网络系统和人工智能应用都会影响用户、企业甚至社会运行。因此，在工程实践中，需要考虑系统的安全性、可靠性、隐私保护、可维护性和可持续发展问题。

例如，在开发信息系统时，不能只考虑功能实现，还要考虑用户数据安全、系统稳定性和法律伦理要求。本专业通过《工程伦理》、专业课程中的案例分析、课程设计和毕业设计等环节，让我们逐步理解工程技术对社会的影响，也培养我们负责任地使用技术的意识。

2. 课程体系

问题 1：你如何理解工程师的责任？

答案：
我认为工程师不仅要掌握技术，更要有社会责任感和工程伦理意识。对于计算机专业来说，我们设计和开发的软件系统、数据系统、网络系统和人工智能应用都

It encompasses programming, along with foundational mathematics, computer systems, algorithm design, databases, networking, operating systems, software engineering, artificial intelligence, and other related fields.

Through project-based teaching and curriculum design, my understanding of the field has undergone a significant transformation. For instance, in the "C Programming" course projects, we not only wrote code but also considered functional design, module division, input/output, exception handling, and debugging; in the "Computer Networks" course, we gained practical understanding of network protocols and system configuration through experiments; and in the "Software Engineering" course, we learned requirements analysis, system design, testing, and documentation writing. These experiences have shown me that the computer science discipline places greater emphasis on systems thinking, engineering practice, and the ability to solve complex problems.

Question 4: What Types of Careers Do You Think Graduates of This Major Can Pursue?

answer：

This program cultivates applied computer engineering and technology professionals. Graduates are qualified to work in the IT industry, research institutions, enterprises, public service departments, and other organizations, engaging in computer-related roles such as software development, system design, web development, mobile application development, database management, algorithm design, data processing and analysis, artificial intelligence application development, system testing, network management, and project management.

With accumulated work experience, graduates can also develop into technical experts, system architects, or project managers.

Question 5: How Do You Understand an Engineer's Responsibilities?

answer：

I believe engineers must not only master technical skills but also possess a strong sense of social responsibility and engineering ethics. In the field of computer science, the software systems, data systems, network systems, and artificial intelligence applications we design and develop significantly impact users, businesses, and even societal operations. Therefore, in engineering practice, it is essential to address issues such as system security, reliability, privacy protection, maintainability, and sustainability.

For example, when developing information systems, one must consider not only functional implementation but also user data security, system stability, and legal and ethical requirements. Through courses such as Engineering Ethics, case analyses in specialized subjects, course projects, and graduation designs, this program enables us to gradually understand the societal impact of engineering technology while fostering our awareness of responsible technology use.

2. Curriculum System

Question 1: How Do You Understand an Engineer's Responsibilities?

answer：

I believe engineers should not only master technical skills but also possess a strong sense of social responsibility and engineering ethics. For computer science professionals, the software systems, data systems, network systems, and artificial intelligence applications we design and develop are all crucial contributions.

会影响用户、企业甚至社会运行。因此，在工程实践中，需要考虑系统的安全性、可靠性、隐私保护、可维护性和可持续发展问题。

例如，在开发信息系统时，不能只考虑功能实现，还要考虑用户数据安全、系统稳定性和法律伦理要求。本专业通过《工程伦理》、专业课程中的案例分析、课程设计和毕业设计等环节，让我们逐步理解工程技术对社会的影响，也培养我们负责任地使用技术的意识。

问题 2：课程体系是否层层递进、逻辑清晰？

答案：
我认为本专业课程体系是比较清晰和递进的，而且这种递进关系也和专业的 11 项毕业学习成果相对应。低年级阶段主要学习高等数学、线性代数、离散数学、Python 程序设计、C 程序设计等课程，重点打牢数学、自然科学和计算机基础，对应培养工程知识和初步的问题分析能力。在此基础上，中年级进一步学习数据结构、数据库原理及应用、计算机网络、计算机组成原理、操作系统、算法设计与分析等专业核心课程，逐步提升我们对复杂计算机系统的理解能力，强化问题分析、研究能力以及现代工具使用能力。

到高年级阶段，课程进一步转向综合应用和工程实践，如软件工程、人工智能、数据挖掘基础、机器学习、智能应用系统开发等课程，以及 C 程序设计课程设计、算法课程设计、软件工程课程设计、智能应用系统开发课程设计、专业实习和毕业设计等实践环节。这些课程和环节更加突出设计/开发解决方案能力、工程与可持续发展意识、工程伦理和职业规范，同时通过团队项目、项目答辩、实习和毕业设计培养我们的团队合作能力、沟通能力和项目管理能力。

此外，终身学习能力贯穿整个培养过程。随着课程难度逐步提升，我们需要不断查阅资料、学习新工具和新技术，特别是在人工智能、大数据、云计算和毕业设计等环节中，自主学习能力要求更高。

因此，整体来看，课程体系不是简单按课程堆叠，而是按照“基础知识—专业核心—综合应用—工程实践—持续发展”的逻辑逐步推进，既体现了课程之间的先后关系，也支撑了 11 项 PLO 的逐层达成。

问题 3：课程体系如何支撑能力培养？

答案：
专业通过三层对齐关系来保证课程体系支撑能力培养。第一，课程与毕业要求支撑矩阵明确每门课程支撑哪些毕业学习成果，并用量化权重表示支撑强度；第二，目标课程矩阵把 11 项毕业要求映射到相关课程模块；第三，毕业要求与培养目标支撑矩阵说明各项学习成果如何支撑长期培养目标。

从学生角度看，这意味着每门课程都有明确的能力培养目标，不同课程共同支撑专业总体培养目标。例如，数学和算法类课程支撑工程知识、问题分析和研究能力；软件工程、数据库、智能应用系统开发等课程支撑系统设计和工程实践能力；

This will affect users, enterprises, and even the functioning of society. Therefore, in engineering practice, it is essential to consider the system's safety, reliability, privacy protection, maintainability, and sustainability.

For example, when developing information systems, one must consider not only functional implementation but also user data security, system stability, and legal and ethical requirements. Through courses such as Engineering Ethics, case analyses in specialized subjects, course projects, and graduation designs, this program enables us to gradually understand the societal impact of engineering technology while fostering our awareness of responsible technology use.

Question 2: Is the Curriculum Systematically Progressive and Logically Clear?

answer：

I believe the curriculum system for this major is well-defined and progressively structured, with this progression aligning closely with the program's 11 graduation learning outcomes. In the early years, students primarily study courses such as Advanced Mathematics, Linear Algebra, Discrete Mathematics, Python Programming, and C Programming, focusing on building a solid foundation in mathematics, natural sciences, and computer science fundamentals, thereby cultivating engineering knowledge and initial problem-solving skills. In the middle years, students advance to core professional courses including Data Structures, Database Principles and Applications, Computer Networks, Computer Organization Principles, Operating Systems, and Algorithm Design and Analysis, progressively enhancing their understanding of complex computer systems while strengthening their problem-solving, research capabilities, and proficiency in using modern tools.

In the upper years, the curriculum shifts further toward integrated applications and engineering practice, encompassing courses such as software engineering, artificial intelligence, fundamentals of data mining, machine learning, and intelligent application system development, along with practical components including C programming course design, algorithm course design, software engineering course design, intelligent application system development course design, professional internships, and graduation projects. These courses and components place greater emphasis on solution design/development capabilities, engineering and sustainability awareness, engineering ethics, and professional standards, while simultaneously fostering teamwork skills, communication abilities, and project management competencies through team projects, project defenses, internships, and graduation projects.

Furthermore, the capacity for lifelong learning permeates the entire training process. As the course difficulty progressively increases, we must continuously consult resources and acquire new tools and technologies, particularly in areas such as artificial intelligence, big data, cloud computing, and graduation projects, where self-directed learning skills are demanded at a higher level.

Therefore, overall, the curriculum system is not merely a simple accumulation of courses, but progresses systematically according to the logic of "basic knowledge—professional core—integrated application—engineering practice—sustainable development." This approach not only reflects the sequential relationship between courses but also facilitates the progressive achievement of the 11 PLOs.

Question 3: How Does the Curriculum System Support Competency Development?

answer：

The program ensures the curriculum system effectively supports competency development through a three-tier alignment framework. First, the Curriculum-Graduation Requirements Support Matrix clearly identifies which graduation learning outcomes each course supports, with quantitative weights indicating the level of support. Second, the Target Course Matrix maps the 11 graduation requirements to corresponding course modules. Third, the Graduation Requirements and Educational Objectives Support Matrix explains how each learning outcome contributes to achieving long-term educational goals.

From the student perspective, this means each course has clearly defined competency objectives, with different courses collectively supporting the program's overall educational goals. For example, mathematics and algorithm courses foster engineering knowledge, problem analysis, and research skills; while courses such as software engineering, databases, and intelligent application system development enhance system design and engineering practice capabilities.

英语、专业英语、科技论文写作等课程支撑沟通和国际交流能力；工程伦理、就业指导、专业实习等课程支撑职业规范、项目管理和社会责任。

问题 4：理论课程和实践课程的平衡如何？

答案：
总体来看，理论课程和实践课程结合比较紧密。理论课程帮助我们理解基本概念、原理和方法，实践课程和课程项目帮助我们把这些知识应用到具体问题中。

例如，程序设计课程配有上机实践，数据库课程有数据库设计与 SQL 操作，计算机网络课程有网络配置与实验，软件工程课程通过项目训练需求分析、设计、编码和测试能力，智能应用系统开发课程则更加注重综合系统开发。专业还有集中实践环节，包括专业见习、课程设计、专业实习、毕业设计和第二课堂。这样的安排让我们既能学到理论，也能通过项目和实验提升实际动手能力。

问题 5：是否存在课程衔接不合理或先修不足的情况？

答案：
整体上课程衔接是比较合理的。培养方案按照基础课程、专业基础课程、专业核心课程、综合实践课程逐步推进。比如，学习数据结构和算法之前，我们已经学习了程序设计和离散数学；学习数据库、操作系统、计算机网络之前，已经具备一定的程序设计和计算机基础；高年级的软件工程、智能应用系统开发、机器学习等课程则建立在前期课程基础上。

个别课程难度较大时，教师会通过复习前置知识、提供学习资料、课后答疑等方式帮助我们过渡。

问题 6：课程是否覆盖本专业核心领域？

答案：
是的。本专业课程覆盖了计算机科学与技术的主要核心领域，包括数学基础、程序设计、数据结构与算法、数字逻辑、计算机组成、操作系统、数据库、计算机网络、软件工程、人工智能、数据挖掘、机器学习、图像处理、Web 与移动应用开发等。

这些课程基本覆盖了计算机系统、软件开发、数据处理、网络技术和智能应用等方向，能够支撑我们未来从事计算机应用领域的工程实践。

问题 7：在人工智能等新兴方向是否有课程或内容？

Courses such as English, Professional English, and Scientific Paper Writing enhance communication and international exchange skills; courses including Engineering Ethics, Career Guidance, and Professional Internships foster professional standards, project management, and social responsibility.

Question 4: How Is the Balance Between Theoretical and Practical Courses Maintained?

answer：

Overall, the theoretical and practical courses are closely integrated. Theoretical courses help us understand fundamental concepts, principles, and methods, while practical courses and project-based learning enable us to apply this knowledge to specific problems.

For example, the programming course includes hands-on computer practice; the database course covers database design and SQL operations; the computer networking course involves network configuration and experiments; the software engineering course develops skills in requirements analysis, design, coding, and testing through project-based training; while the intelligent application system development course places greater emphasis on comprehensive system development. The program also features concentrated practical components, including professional internships, course projects, professional practicums, graduation designs, and extracurricular activities. This structure enables students to acquire both theoretical knowledge and enhance their practical skills through projects and experiments.

Question 5: Are There Any Issues With Unreasonable Course Sequencing or Insufficient Prerequisite Requirements?

answer：

Overall, the course progression is well-structured. The training program advances progressively through foundational courses, professional foundation courses, core professional courses, and comprehensive practical courses. For instance, before studying data structures and algorithms, students have already covered programming and discrete mathematics; prior to learning databases, operating systems, and computer networks, they possess a solid foundation in programming and computer basics; advanced courses such as software engineering, intelligent application system development, and machine learning are built upon this earlier curriculum.

When individual courses are particularly challenging, instructors assist students in navigating the transition by reviewing prerequisite knowledge, providing learning materials, and offering post-class consultations.

Question 6: Does the Course Cover the Core Areas of This Discipline?

answer：

Yes. This program covers the core areas of computer science and technology, including mathematical foundations, programming, data structures and algorithms, digital logic, computer architecture, operating systems, databases, computer networks, software engineering, artificial intelligence, data mining, machine learning, image processing, and Web and mobile application development.

These courses comprehensively cover areas such as computer systems, software development, data processing, network technology, and intelligent applications, equipping students with the foundation for future engineering practice in computer application fields.

Question 7: Are There Any Courses or Content in Emerging Fields Such as Artificial Intelligence?

答案：
有。专业课程中已经包含人工智能相关内容，例如《人工智能》《机器学习》《数据挖掘基础》《智能控制原理》《人机交互技术》《智能应用系统开发》《机器人技术与应用》《AI 大模型应用开发》《数字图像处理》《数字孪生技术》《医疗大数据》《智慧医疗系统》等课程。同时，一些课程项目也会结合大数据、机器学习、智能应用系统和大模型 API 等新技术。

此外，学院也通过实验平台、教师科研项目、学科竞赛和第二课堂活动，让学生有机会接触人工智能和数据智能方向的实际应用。

问题 8：课程内容是否与工程实践结合？

答案：
课程内容与工程实践结合比较紧密。很多课程都安排了实验、项目或课程设计。例如，《C 程序设计》通过小型管理系统训练编程能力；《数据库原理及应用》通过数据库建模和查询训练数据管理能力；《软件工程》通过团队项目训练需求分析、系统设计和文档写作；《智能应用系统开发》要求我们完成较完整的应用系统设计和开发。

一些项目来源于企业需求、教师科研项目或实际应用场景，企业工程师也会通过讲座、实习指导或项目评审参与人才培养，使课程内容更贴近工程实际。

问题 9：学校提供哪些选修课程？是否丰富？

答案：
学校提供了较丰富的选修课程，主要包括人工智能、软件开发、大数据、云计算、网络安全、智慧医疗、Web 应用开发、Hadoop、Spark、大数据开发技术、AI 大模型应用开发、科技论文写作等课程。

这些课程可以支持不同兴趣方向的学生发展。例如，对软件开发感兴趣的学生可以选择 Web 应用开发、移动应用开发、Java 进阶等课程；对人工智能和数据方向感兴趣的学生可以选择机器学习、数据挖掘、大数据技术等课程；对安全方向感兴趣的学生可以选择网络安全课程。

问题 10：选修课程是否支持个性化发展？

答案：
支持。学生可以根据兴趣、职业规划和学习基础选择不同方向的课程。除了课堂选修课程，学校还提供第二课堂、学科竞赛、创新创业训练、科研训练和企业实践等机会。学生可以根据自己的发展目标选择偏软件开发、人工智能、大数据、网络安全或项目管理等方向，逐步形成自己的能力特色。

answer：

Yes. The professional curriculum already incorporates content related to artificial intelligence, including courses such as "Artificial Intelligence," "Machine Learning," "Fundamentals of Data Mining," "Principles of Intelligent Control," "Human-Computer Interaction Technology," "Intelligent Application System Development," "Robotics Technology and Applications," "AI Large Model Application Development," "Digital Image Processing," "Digital Twin Technology," "Medical Big Data," and "Smart Healthcare Systems." Additionally, certain course projects integrate cutting-edge technologies like big data, machine learning, intelligent application systems, and large model APIs.

Furthermore, the college provides students with opportunities to engage with practical applications in artificial intelligence and data intelligence through experimental platforms, faculty research projects, academic competitions, and extracurricular activities.

Question 8: Is the Course Content Integrated With Engineering Practice?

answer：

The course content is closely integrated with engineering practice. Many courses include experiments, projects, or course designs. For example, "C Programming" enhances programming skills through small-scale management system exercises; "Database Principles and Applications" develops data management capabilities via database modeling and query techniques; "Software Engineering" cultivates skills in requirements analysis, system design, and documentation writing through team projects; while "Intelligent Application System Development" requires students to complete comprehensive application system design and development.

Some projects stem from corporate needs, faculty research initiatives, or real-world application scenarios. Corporate engineers also contribute to talent development through lectures, internship guidance, or project reviews, ensuring the curriculum content closely aligns with practical engineering requirements.

Question 9: What Elective Courses Does the School Offer? Are They Diverse?

answer：

The university offers a comprehensive range of elective courses, including Artificial Intelligence, Software Development, Big Data, Cloud Computing, Cybersecurity, Smart Healthcare, Web Application Development, Hadoop, Spark, Big Data Development Technologies, AI Large Model Application Development, and Scientific Paper Writing.

These courses cater to students with diverse interests. For instance, those interested in software development can choose courses such as Web Application Development, Mobile Application Development, and Advanced Java; students interested in artificial intelligence and data may opt for courses like Machine Learning, Data Mining, and Big Data Technology; while those focused on security can select cybersecurity courses.

Question 10: Do Elective Courses Support Personalized Development?

answer：

Support. Students can choose courses in various directions based on their interests, career aspirations, and academic foundation. In addition to elective courses, the school offers opportunities such as extracurricular activities, academic competitions, innovation and entrepreneurship training, research training, and corporate internships. Students can select specializations in software development, artificial intelligence, big data, cybersecurity, or project management according to their personal development goals, gradually cultivating their unique competencies.

问题 11：你是否能根据兴趣构建自己的知识体系？

答案：
可以。基础课程和专业核心课程帮助我们建立共同的专业基础，选修课程、课程项目、竞赛和实习则提供个性化发展空间。例如，如果我对人工智能感兴趣，可以重点学习 Python、数据结构、算法、人工智能、机器学习、数据挖掘和智能应用系统开发；如果对软件开发感兴趣，可以重点学习数据库、软件工程、Java、Web 应用开发和移动应用开发。

通过课程、实践和竞赛结合，可以逐步形成自己的知识体系和发展方向。

问题 12：你认为所学课程是否有用？

答案：
我认为课程整体是有用的。基础课程帮助我们掌握数学、程序设计和计算机基本原理；专业课程帮助我们理解系统、网络、数据库、算法和软件工程；实践课程帮助我们把知识用于实际系统开发。

例如，《数据结构》和《算法设计与分析》提升了我们分析问题和设计程序的能力；《数据库原理及应用》在很多项目中都会用到；《软件工程》让我们理解真实软件开发过程；《计算机网络》帮助我们理解网络系统运行和配置。这些课程对就业、实习和毕业设计都有帮助。

问题 13：项目化教学对你有什么帮助？

答案：
项目化教学最大的好处是让我们不只是“学知识点”，而是学会把知识真正用起来。比如在《C 程序设计》《数据库原理及应用》《软件工程》《智能应用系统开发》等课程中，我们需要围绕一个具体项目完成需求分析、功能设计、代码实现、测试调试、文档撰写和最终汇报。这个过程让我们更清楚地理解课程内容在实际工程中的作用。

项目化教学也明显提升了我们的工程实践能力和问题解决能力。做项目时，经常会遇到课本上没有直接答案的问题，比如程序报错、数据库设计不合理、接口调用失败、系统功能无法正常运行等。我们需要自己查资料、调试代码、和同学讨论，必要时再向老师请教。通过这个过程，我们的问题分析能力、现代工具使用能力和自主学习能力都有明显提升。

另外，项目通常是小组完成的，所以也锻炼了团队协作和沟通能力。我们需要分工、协调进度、整合代码、准备展示和答辩，这和真实的软件开发流程比较接近。通过项目化教学，我们不仅掌握了专业知识，也提前熟悉了工程项目的基本流程，对以后实习、毕业设计和就业都有帮助。

Question 11: Can You Build Your Own Knowledge System Based on Your Interests?

answer：

Certainly. The foundational courses and core professional courses help us build a shared professional foundation, while elective courses, course projects, competitions, and internships provide opportunities for personalized development. For example, if I am interested in artificial intelligence, I can focus on Python, data structures, algorithms, artificial intelligence, machine learning, data mining, and intelligent application system development; if I am interested in software development, I can concentrate on databases, software engineering, Java, web application development, and mobile application development.

By integrating courses, practical experience, and competitions, one can gradually build a personalized knowledge system and define their career path.

Question 12: Do You Think the Courses You Have Taken Are Useful?

answer：

I believe the entire curriculum is highly effective. The foundational courses equip us with a solid grasp of mathematics, programming, and fundamental computer principles; the specialized courses deepen our understanding of systems, networks, databases, algorithms, and software engineering; while the practical courses enable us to apply this knowledge to real-world system development.

For example, Data Structures and Algorithm Design and Analysis enhance our problem-solving and programming skills; Database Principles and Applications is widely used in various projects; Software Engineering provides insight into the actual software development process; Computer Networks helps us understand how network systems operate and are configured. These courses are beneficial for career development, internships, and graduation projects.

Question 13: How Has Project-Based Teaching Benefited You?

answer：

The greatest advantage of project-based teaching is that it enables us not merely to "learn knowledge points," but to truly apply that knowledge. For instance, in courses such as "C Programming," "Database Principles and Applications," "Software Engineering," and "Intelligent Application System Development," we complete a series of tasks centered around a specific project: requirements analysis, functional design, code implementation, testing and debugging, documentation preparation, and final presentation. This process provides a clearer understanding of how the course content is applied in real-world engineering contexts.

Project-based teaching has significantly enhanced our engineering practice skills and problem-solving abilities. When working on projects, we frequently encounter problems without direct solutions in textbooks, such as program errors, poorly designed databases, failed interface calls, or malfunctioning system functions. We must independently research materials, debug code, discuss with classmates, and seek guidance from instructors when necessary. Through this process, our problem analysis skills, proficiency in using modern tools, and self-learning capabilities have all improved markedly.

Moreover, projects are typically completed by teams, thereby enhancing teamwork and communication skills. The process involves task division, schedule coordination, code integration, and preparation for presentations and defenses—all closely mirroring real-world software development workflows. Through project-based learning, we not only acquire professional knowledge but also gain early familiarity with fundamental engineering project procedures, which proves beneficial for future internships, graduation projects, and career development.

问题 14：每门课程需要投入多少课外时间？

答案：
不同课程需要投入的课外时间不太一样，主要和课程性质、学分、实验项目和考核方式有关。

从我们的实际学习来看，一般课程通常是 1 个学分对应 16 个课内学时和 14 个课外学习时间。

课外学习时间主要用于几个方面：第一，完成课后线上线下作业和线上编程练习，比如程序设计、数据结构、算法等课程需要自己写代码、调试程序；第二，完成实验或上机任务，比如数据库、计算机网络、Python 程序设计等课程需要做实验报告、配置环境或测试结果；第三，进行线上线下自主复习和预习，特别是操作系统、计算机组成原理、算法设计与分析等课程，课后需要花时间理解概念和整理笔记；第四，完成课程项目、课程设计和小组任务，例如软件工程课程设计、智能应用系统开发课程设计需要进行需求分析、系统设计、代码实现、测试和文档撰写；第五，进行考试准备，包括复习重点、做练习题、整理错题和准备课程答辩。

在项目类课程和毕业设计阶段，课外投入会更多。比如软件工程课程设计需要小组讨论、分工协作、撰写文档和系统测试；算法课程设计需要查找资料、设计算法并完成程序实现；智能应用系统开发课程设计需要综合使用数据库、前后端、人工智能或接口调用等技术；毕业设计则需要长期投入文献阅读、系统开发、论文撰写和答辩准备。

总体来说，课外学习时间不是简单“复习一下”，而是包括作业、实验、项目开发、资料查阅、工具学习、报告撰写和考试准备等完整学习过程。学校和学院也会通过 workload 调查、课程反馈和教学评价来了解我们实际投入的学习时间，如果发现课程负担不均衡，会在课程内容、作业安排或课程进度上进行调整。

问题 15：学习负担是否均衡？

答案：
整体上学习负担是可以接受的，课程安排大体符合学习规律。不过在专业课程较集中、课程设计较多的学期，负担会明显增加。例如同时有专业核心课程、实验、课程项目和考试时，需要较强的时间管理能力。

学校也通过课程安排、教学反馈、学生 workload 调查等方式了解学生学习负担，并在培养方案修订和课程安排中进行调整。

问题 16：是否存在特别紧张的学期？

Question 14: How Much Extracurricular Time Is Required for Each Course?

answer：

The amount of extracurricular time required varies across different courses, primarily depending on the course's nature, credit requirements, laboratory components, and assessment methods.

Based on our practical experience, a standard course typically consists of 1 credit equivalent to 16 instructional hours and 14 extracurricular study hours.

Extracurricular study time is primarily allocated to the following areas: First, completing after-class assignments both online and offline, as well as online programming exercises—such as writing code and debugging programs required in courses like programming, data structures, and algorithms; Second, executing laboratory experiments or hands-on tasks, including preparing lab reports, configuring development environments, or verifying results for courses like databases, computer networks, and Python programming; Third, conducting self-directed review and previewing material both online and offline, particularly for courses such as operating systems, computer architecture principles, and algorithm design and analysis, where students need to thoroughly understand concepts and organize notes post-class; Fourth, completing course projects, design assignments, and group tasks—including requirement analysis, system design, code implementation, testing, and documentation preparation for software engineering and intelligent application system development courses; Fifth, preparing for examinations through reviewing key topics, practicing exercises, organizing error logs, and preparing for course defenses.

During project-based courses and the graduation design phase, students invest significantly more time and effort outside the classroom. For instance, software engineering course projects require group discussions, collaborative task division, documentation preparation, and system testing; algorithm course projects involve literature research, algorithm design, and program implementation; intelligent application system development projects necessitate the integrated application of technologies such as databases, front-end/back-end development, artificial intelligence, and API calls; while graduation design demands sustained effort in literature review, system development, thesis writing, and defense preparation.

Overall, extracurricular learning time is not merely about "reviewing material," but encompasses a comprehensive learning process that includes homework, experiments, project development, literature review, tool acquisition, report writing, and exam preparation. Schools and colleges also assess the actual learning time invested through workload surveys, course feedback, and teaching evaluations. If an imbalance in course workload is identified, adjustments are made to course content, assignment distribution, or course pacing.

Question 15: Is the Academic Workload Balanced?

answer：

Overall, the academic workload is acceptable, and the course schedule generally aligns with learning principles. However, during semesters with a high concentration of specialized courses and numerous course assignments, the workload increases significantly. For instance, when core professional courses, laboratory sessions, course projects, and examinations are scheduled concurrently, strong time management skills are required.

The school also assesses students' academic workload through course scheduling, teaching feedback, and student workload surveys, making corresponding adjustments in curriculum revisions and course arrangements.

Question 16: Is There a Particularly Stressful Semester?

答案：
从大二到大三阶段会比较紧张，因为专业核心课程和课程设计较多。例如数据结构、数据库、计算机网络、操作系统、算法设计与分析、软件工程等课程都需要较多课后学习和实践。到大四阶段，专业实习和毕业设计也需要持续投入。

不过这些阶段也是能力提升最快的阶段，只要合理安排时间，基本可以完成学习任务。

问题 17：学校提供了哪些实践机会？

答案：
学校提供了比较完整的实践教学体系，包括专业见习、C 程序设计课程设计、算法设计与分析课程设计、软件工程课程设计、智能应用系统开发课程设计、专业实习、毕业设计和第二课堂素质拓展。

此外，课程中还有实验和项目训练，学校也组织企业参观、企业讲座、学科竞赛、创新创业项目和科研训练。通过这些实践机会，我们可以逐步从基础编程训练过渡到系统设计和真实工程问题解决。

问题 18：专业见习有什么作用？

答案：
专业见习主要帮助我们了解行业、企业岗位和工程流程。通过企业参观、工程师讲座、行业交流等方式，我们可以在正式进入专业深度学习之前，对计算机行业、企业开发流程和岗位能力要求有基本认识。

它对我们明确学习方向、理解课程意义和规划职业发展都有帮助。

问题 19：C 程序设计课程设计如何实施？题目是谁定？可以自己选任务吗？

答案：
C 程序设计课程设计一般安排在大一或大二早期，主要目的是巩固 C 语言基础，提高编程能力和问题解决能力。题目通常由指导教师统一设计并发布，包含若干不同类型和难度的任务，例如学生成绩管理系统、图书管理系统、考勤管理系统、简易计算器等。

学生可以在教师提供的题目列表中根据兴趣和能力选择任务，也可以在教师同意后对题目进行适当调整。完成过程中需要进行功能设计、代码编写、调试测试和报告撰写。

answer：

The transition from sophomore to junior year is particularly demanding due to the heavy workload of core professional courses and project-based assignments. Courses such as Data Structures, Database Systems, Computer Networks, Operating Systems, Algorithm Design and Analysis, and Software Engineering require substantial self-study and hands-on practice. In the senior year, students must also dedicate consistent effort to professional internships and graduation projects.

However, these stages are also when skills improve most rapidly. With proper time management, learning objectives can generally be achieved.

Question 17: What Practical Opportunities Does the School Offer?

answer：

The university offers a comprehensive practical teaching system, encompassing professional internships, C programming course projects, algorithm design and analysis course projects, software engineering course projects, intelligent application system development course projects, professional internships, graduation projects, and extracurricular quality enhancement activities.

In addition, the curriculum includes laboratory experiments and project-based training. The school also organizes corporate visits, industry lectures, academic competitions, innovation and entrepreneurship projects, and research training. Through these practical opportunities, students can progressively transition from fundamental programming skills to system design and the resolution of real-world engineering challenges.

Question 18: What Is the Purpose of Professional Internship?

answer：

Professional internships primarily assist us in gaining an understanding of the industry, corporate roles, and engineering workflows. Through methods such as company visits, engineer lectures, and industry exchanges, we can acquire a fundamental knowledge of the computer industry, corporate development processes, and job competency requirements before embarking on in-depth professional studies.

It helps us clarify our learning direction, understand the significance of the courses, and plan our career development.

Question 19: How Is the C Programming Course Design Implemented? Who Sets the Topics? Can Students Choose Their Own Tasks?

answer：

The C Programming course design project is typically scheduled during the first or early second year of university, aiming primarily to reinforce C language fundamentals and enhance programming and problem-solving skills. The projects are uniformly designed and assigned by instructors, encompassing various tasks of different types and difficulty levels, such as student grade management systems, library management systems, attendance management systems, and basic calculators.

Students can select tasks from the list provided by the teacher based on their interests and abilities, or make appropriate adjustments to the tasks with the teacher's approval. The completion process involves functional design, code writing, debugging and testing, and report writing.

问题 20：算法设计与分析课程设计主要训练什么能力？

答案：
算法课程设计主要训练问题抽象、算法设计、复杂度分析和程序实现能力。我们需要针对具体问题选择或设计合适算法，并通过编程实现和测试验证算法效果。

这个过程不仅帮助我们理解课堂上的算法思想，也训练我们将理论方法用于实际计算问题的能力，是专业核心能力的重要支撑。

问题 21：软件工程课程设计如何体现工程化开发？

答案：
软件工程课程设计通常按照较完整的软件开发流程进行，包括需求分析、概要设计、详细设计、编码实现、测试、文档撰写和项目答辩。学生一般以小组形式完成项目，每个人承担不同角色和任务。

通过这个过程，我们能够学习如何进行团队协作、任务分解、进度管理和质量控制，也能理解软件开发不仅是写代码，还包括需求、设计、测试、文档和项目管理等环节。

问题 22：智能应用系统开发课程设计有什么特点？

答案：
智能应用系统开发课程设计更强调综合开发和新技术应用。它通常要求我们综合使用前期学习的程序设计、数据库、算法、软件工程和人工智能相关知识，完成一个具有一定实际应用价值的系统。

项目可能涉及数据处理、模型调用、接口设计、前后端开发、系统集成和部署等内容。这类课程设计能够提升我们综合应用能力和创新意识。

问题 23：专业实习如何开展？对学生有什么帮助？

答案：
专业实习一般安排在高年级阶段，学生进入企业或相关单位参加真实岗位实践。实习内容可能包括软件开发、系统测试、数据处理、运维支持、文档撰写等。

专业实习帮助我们了解企业真实工作流程、项目规范和岗位要求，也让我们把校内学习的知识转化为实际工作能力。通过企业导师和校内导师共同指导，我们可以更好地适应未来就业环境。

问题 24：第二课堂素质拓展包括哪些内容？

Question 20: What Core Competencies Does the Algorithm Design and Analysis Course Primarily Cultivate?

answer：

The algorithm course design primarily cultivates skills in problem abstraction, algorithm design, complexity analysis, and program implementation. Students are required to select or design appropriate algorithms for specific problems, then implement them through programming and validate their performance through testing.

This process not only helps us understand the algorithmic thinking in the classroom but also trains our ability to apply theoretical methods to practical computational problems, serving as a crucial foundation for core professional competencies.

Question 21: How Does the Software Engineering Course Design Reflect Engineering Development Practices?

answer：

The software engineering course design typically follows a comprehensive software development lifecycle, encompassing requirements analysis, preliminary design, detailed design, coding implementation, testing, documentation preparation, and project defense. Students generally complete the project in groups, with each member assuming distinct roles and responsibilities.

Through this process, we learn how to collaborate as a team, break down tasks, manage timelines, and ensure quality control. We also understand that software development involves more than just writing code—it encompasses requirements gathering, design, testing, documentation, and project management.

Question 22: What Are the Distinctive Features of the Intelligent Application System Development Course Design?

answer：

The Intelligent Application System Development course emphasizes comprehensive development and the application of new technologies. It typically requires students to integrate knowledge acquired earlier—from programming, databases, algorithms, software engineering, and artificial intelligence—to create a system with practical application value.

The project may involve data processing, model invocation, interface design, front-end and back-end development, system integration, and deployment. Such course projects enhance our comprehensive application skills and innovative mindset.

Question 23: How Is the Professional Internship Conducted? What Benefits Does It Offer to Students?

answer：

Professional internships are typically arranged during the senior years, when students join enterprises or relevant organizations to engage in hands-on practical experience in real-world positions. Internship responsibilities may include software development, system testing, data processing, operations and maintenance support, and documentation preparation.

Professional internships enable us to understand the actual workflow, project standards, and job requirements within enterprises, while also facilitating the transformation of academic knowledge into practical work competencies. Under the joint guidance of corporate mentors and academic supervisors, we can better adapt to future employment environments.

Question 24: What Does Extracurricular Quality Development Include?

答案：
第二课堂包括学科竞赛、创新创业训练、科研训练、技术讲座、社团活动、志愿服务和职业素养训练等。它补充了课内教学，帮助学生提升创新能力、团队协作能力、沟通表达能力和综合素养。

对于计算机专业学生来说，第二课堂中的编程竞赛、软件设计竞赛、创新创业项目和科研训练，对提升实践能力和就业竞争力都很有帮助。

问题 25：毕业设计或毕业论文如何帮助你综合运用知识？

答案：
毕业设计是对大学四年学习成果的综合检验。学生需要在导师指导下完成选题、调研、方案设计、系统实现或研究分析、测试验证、论文或设计说明书撰写和答辩。

对于计算机专业来说，很多毕业设计题目需要开发一个系统或完成一个应用研究，例如 Web 系统、移动应用、数据分析平台或智能应用系统。这个过程能够综合运用程序设计、数据库、软件工程、算法和专业方向知识，也训练文献查阅、工程实现、问题分析和表达汇报能力。

3. 教学资源

问题 1：实验课一般多少人？配备多少教师或指导人员？

答案：
实验课一般按照班级或小班进行，具体人数与课程安排有关。实验过程中通常由任课教师负责指导，有些课程还会配备实验教师或助教。对于项目类课程，教师会在课堂和课后通过线上平台、答疑时间或小组指导提供支持。

问题 2：课程大纲是否公开？学生如何查看？

答案：
课程大纲对学生是公开的。学生可以通过教学管理系统、课程平台或任课教师提供的资料查看课程大纲。教师通常也会在第一次课介绍课程目标、教学内容、考核方式、学习要求和课程对毕业学习成果的支撑关系。

问题 3：课程大纲中是否说明课程目标和考核方式？

answer：

The second classroom encompasses academic competitions, innovation and entrepreneurship training, research training, technical lectures, club activities, volunteer services, and professional competency training. It complements classroom instruction by helping students enhance their innovative capabilities, teamwork skills, communication abilities, and overall competence.

For computer science students, extracurricular activities such as programming competitions, software design contests, innovation and entrepreneurship projects, and research training are highly beneficial for enhancing practical skills and boosting employability.

Question 25: How Does Your Graduation Project or Thesis Help You Apply Knowledge Comprehensively?

answer：

The graduation project serves as a comprehensive evaluation of the learning outcomes achieved over four years of university study. Under the guidance of their supervisors, students are required to complete the following tasks: topic selection, research, proposal design, system implementation or analysis, testing and validation, writing of the thesis or design report, and defense presentation.

For computer science majors, many graduation project topics involve developing a system or conducting applied research, such as web systems, mobile applications, data analysis platforms, or intelligent application systems. This process enables the integrated application of programming, database technology, software engineering, algorithms, and specialized domain knowledge, while also cultivating skills in literature review, engineering implementation, problem analysis, and presentation and reporting abilities.

3. Teaching Materials

Question 1: How Many Students Typically Attend an Experimental Course? How Many Instructors or Supervisors Are Assigned?

answer：

Laboratory sessions are typically conducted by class or small groups, with the exact number of participants determined by the course schedule. During experiments, instructors generally provide guidance, and some courses may also include laboratory instructors or teaching assistants. For project-based courses, instructors offer support both in class and after class through online platforms, Q&A sessions, or group guidance.

Question 2: Is the Course Syllabus Public? How Can Students View It?

answer：

The course syllabus is publicly available to students. Students can access the syllabus through the teaching management system, the course platform, or materials provided by the instructor. Typically, instructors also outline the course objectives, teaching content, assessment methods, learning requirements, and the course's contribution to graduate learning outcomes during the first lecture.

Question 3: Does the Course Syllabus Specify the Course Objectives and Assessment Methods?

答案：
是的。课程大纲通常包括课程目标、教学内容、教学方法、考核方式、成绩构成、课程支撑的毕业要求指标点等内容。这样学生可以在课程开始时了解本课程要学什么、如何考核以及课程对能力培养的作用。

4. 学生支持与服务

问题 1：你是否能根据兴趣构建自己的知识体系？

答案：
可以。基础课程和专业核心课程帮助我们建立共同的专业基础，选修课程、课程项目、竞赛和实习则提供个性化发展空间。例如，如果我对人工智能感兴趣，可以重点学习 Python、数据结构、算法、人工智能、机器学习、数据挖掘和智能应用系统开发；如果对软件开发感兴趣，可以重点学习数据库、软件工程、Java、Web 应用开发和移动应用开发。

通过课程、实践和竞赛结合，可以逐步形成自己的知识体系和发展方向。

问题 2：你认为所学课程是否有用？

答案：
我认为课程整体是有用的。基础课程帮助我们掌握数学、程序设计和计算机基本原理；专业课程帮助我们理解系统、网络、数据库、算法和软件工程；实践课程帮助我们把知识用于实际系统开发。

例如，《数据结构》和《算法设计与分析》提升了我们分析问题和设计程序的能力；《数据库原理及应用》在很多项目中都会用到；《软件工程》让我们理解真实软件开发过程；《计算机网络》帮助我们理解网络系统运行和配置。这些课程对就业、实习和毕业设计都有帮助。

问题 3：竞赛是否增加学生负担？如何保障学生权益？

答案：
竞赛参与坚持自愿原则，不强制要求学生参加。学生可以根据自己的兴趣、能力和发展规划选择是否参加。学校和学院通常提供竞赛指导、训练资源、场地和平台支持，也通过创新学分、综合测评或奖励机制鼓励学生参与。

同时，指导教师会关注学生的学习负担和心理状态，帮助学生合理安排课程学习和竞赛训练。竞赛的目的是促进学生成长，而不是给学生增加不必要压力。

answer：
Yes. The course syllabus typically includes course objectives, teaching content, instructional methods, assessment methods, grade composition, and the graduation requirement indicators supported by the course. This enables students to understand at the outset what the course covers, how it is assessed, and its role in fostering competencies.

4. Student Support and Services

Question 1: Can You Build Your Own Knowledge System Based on Your Interests?

answer：
Certainly. The foundational courses and core professional courses help us build a shared professional foundation, while elective courses, course projects, competitions, and internships provide opportunities for personalized development. For example, if I am interested in artificial intelligence, I can focus on Python, data structures, algorithms, artificial intelligence, machine learning, data mining, and intelligent application system development; if I am interested in software development, I can concentrate on databases, software engineering, Java, web application development, and mobile application development.

By integrating courses, practical experience, and competitions, one can gradually build a personalized knowledge system and define their career path.

Question 2: Do You Think the Courses You Have Taken Are Useful?

answer：
I believe the entire curriculum is highly effective. The foundational courses equip us with a solid grasp of mathematics, programming, and fundamental computer principles; the specialized courses deepen our understanding of systems, networks, databases, algorithms, and software engineering; while the practical courses enable us to apply this knowledge to real-world system development.

For example, Data Structures and Algorithm Design and Analysis enhance our problem-solving and programming skills; Database Principles and Applications is widely used in various projects; Software Engineering provides insight into the actual software development process; Computer Networks helps us understand how network systems operate and are configured. These courses are beneficial for career development, internships, and graduation projects.

Question 3: Does the Competition Increase Students 'burden? How Can Students' Rights Be Protected?

answer：
Competition participation adheres to the principle of voluntariness and is not mandatory for students. Students may choose to participate based on their interests, abilities, and developmental plans. Schools and colleges typically provide competition guidance, training resources, venue and platform support, and encourage student participation through innovative credit systems, comprehensive evaluation mechanisms, or reward programs.

Meanwhile, instructors will monitor students' academic workload and psychological well-being, assisting them in effectively balancing coursework and competition training. The purpose of competitions is to foster student development rather than impose unnecessary stress.

问题 4：参加竞赛对你有什么帮助？

答案：
参加竞赛可以提升工程实践能力、创新能力、团队协作能力和表达能力。在竞赛中，我们通常要面对开放性问题，需要查资料、设计方案、编程实现、测试优化并进行答辩展示。

这些过程和真实工程项目比较接近，能够锻炼综合能力。竞赛成果还可以与课程设计、毕业设计或创新创业项目结合，对就业和深造也有帮助。

问题 5：学校是否提供创业支持？

答案：
学校提供创新创业课程、创新创业训练项目、竞赛指导和创业咨询等支持。如果学生有创业想法，可以通过创新创业课程、就业指导与创业基础课程、学院教师指导、创新创业项目申报、创业竞赛等渠道获得帮助。

问题 6：你们课后如何联系教师？

答案：
可以通过课堂答疑、办公室答疑时间、课程 QQ 群或微信群、学习平台、邮件等方式联系教师。有些课程教师会在超星等在线平台发布资料、作业和通知，学生也可以在线提问。

问题 7：如果学习遇到困难，学院有什么帮助？

答案：
学院有比较系统的学生支持机制。任课教师会提供课后答疑，课程平台提供学习资料，班主任和辅导员会关注学生学习状态。对于学业困难学生，学院会通过学业预警、谈话帮扶、学习指导、优秀学生帮扶等方式帮助学生改进学习。

问题 8：学生如果对考试成绩有疑问，可以怎么办？

答案：
学生可以按照学校规定申请成绩复核。一般流程是学生向所在学院提交书面申请，经学院教学负责人审核后，由学院或教务处组织专人复核试卷或成绩记录。如果发现评分、登分或统计错误，会按规定程序进行更正。该机制能够保障成绩评定的公平性和规范性。

Question 4: What Benefits Does Participating in the Competition Bring You?

answer：

Participating in competitions enhances practical engineering skills, innovation capabilities, teamwork abilities, and communication skills. During these events, participants typically address open-ended questions, involving research, design proposals, programming implementation, testing and optimization, followed by a presentation and defense session.

These processes closely resemble real-world engineering projects, enabling the development of comprehensive skills. The outcomes of these competitions can also be integrated with course designs, graduation projects, or innovation and entrepreneurship initiatives, thereby enhancing career prospects and further academic pursuits.

Question 5: Does the School Provide Entrepreneurial Support?

answer：

The university provides support including innovation and entrepreneurship courses, innovation and entrepreneurship training programs, competition guidance, and entrepreneurial consulting. Students with entrepreneurial ideas can receive assistance through various channels such as innovation and entrepreneurship courses, career guidance and foundational entrepreneurship courses, mentorship from faculty members, application for innovation and entrepreneurship projects, and entrepreneurial competitions.

Question 6: How Do You Contact Your Teacher After Class?

answer：

Students can contact instructors through classroom Q&A sessions, office consultation hours, course-specific QQ/WeChat groups, learning platforms, or email. For certain courses, instructors release materials, assignments, and announcements on online platforms like Chaoxing, where students can also submit questions online.

Question 7: What Assistance Does the College Provide if Students Encounter Difficulties in Their Studies?

answer：

The college has a relatively systematic student support mechanism. Course instructors provide after-class tutoring, the course platform offers learning materials, and class advisors and counselors monitor students' academic progress. For students facing academic difficulties, the college offers assistance through academic early warning systems, counseling sessions, study guidance, and support programs for high-performing students to help them improve their academic performance.

Question 8: What Should Students Do if They Have Questions About Their Exam Results?

answer：

Students may apply for grade review in accordance with school regulations. The standard procedure involves submitting a written application to their respective college; after review by the college's academic affairs officer, a designated staff member from the college or the Academic Affairs Office will verify the exam papers or grade records. Any errors in grading, score recording, or statistical calculations will be corrected following established procedures. This mechanism ensures the fairness and standardization of the grade evaluation process.

问题 9：如果学生学习遇到困难，学院有什么帮扶机制？

答案：
学院建立了多层次学生支持机制。课程层面，任课教师提供课后答疑和学习平台资源；班级层面，班主任和辅导员关注学生学习状态；学院层面，建立学业预警和帮扶制度，对学分不足、课程不及格较多或学习困难学生进行提醒、谈话和帮扶；同时，可以安排任课教师、班主任或优秀学生进行一对一学习指导。
此外，课程网站和在线教学平台提供学习资料、作业反馈和讨论渠道，帮助学生进行自主学习。

5. 质量保障

问题 1：是否存在课程衔接不合理或先修不足的情况？

答案：
整体上课程衔接是比较合理的。培养方案按照基础课程、专业基础课程、专业核心课程、综合实践课程逐步推进。比如，学习数据结构和算法之前，我们已经学习了程序设计和离散数学；学习数据库、操作系统、计算机网络之前，已经具备一定的程序设计和计算机基础；高年级的软件工程、智能应用系统开发、机器学习等课程则建立在前期课程基础上。

个别课程难度较大时，教师会通过复习前置知识、提供学习资料、课后答疑等方式帮助我们过渡。

问题 2：你认为所学课程是否有用？

答案：
我认为课程整体是有用的。基础课程帮助我们掌握数学、程序设计和计算机基本原理；专业课程帮助我们理解系统、网络、数据库、算法和软件工程；实践课程帮助我们把知识用于实际系统开发。

例如，《数据结构》和《算法设计与分析》提升了我们分析问题和设计程序的能力；《数据库原理及应用》在很多项目中都会用到；《软件工程》让我们理解真实软件开发过程；《计算机网络》帮助我们理解网络系统运行和配置。这些课程对就业、实习和毕业设计都有帮助。

问题 3：你如何评价任课教师的教学质量？

答案：
从学生角度来看，我认为任课教师的整体教学质量是比较好的。大多数教师备课认真，对课程内容比较熟悉，也能按照课程大纲组织教学，让我们清楚这门课的学习目标、主要内容、考核方式以及它和专业毕业学习成果之间的关系。比如在一些专业核心课程中，教师不仅讲授理论知识，也会结合案例、实验或项目帮

Question 9: What Support Mechanisms Does the College Have for Students Facing Learning Difficulties?

answer：

The college has established a multi-tiered student support system. At the course level, instructors provide post-class Q&A sessions and learning platform resources; at the class level, class advisors and counselors monitor students' academic progress; at the college level, an academic early-warning and support system is in place, offering reminders, counseling sessions, and assistance to students with insufficient credits, high failure rates, or learning difficulties. Additionally, one-on-one academic guidance can be arranged by instructors, class advisors, or outstanding students.

In addition, the course website and online teaching platform provide learning materials, assignment feedback, and discussion forums to support students in their self-directed learning.

5. QA

Question 1: Are There Any Issues With Improper Course Sequencing or Insufficient Prerequisite Requirements?

answer：

Overall, the course progression is well-structured. The training program advances progressively through foundational courses, professional foundation courses, core professional courses, and comprehensive practical courses. For instance, before studying data structures and algorithms, students have already covered programming and discrete mathematics; prior to learning databases, operating systems, and computer networks, they possess a solid foundation in programming and computer basics; advanced courses such as software engineering, intelligent application system development, and machine learning are built upon this earlier curriculum.

When individual courses are particularly challenging, instructors assist students in navigating the transition by reviewing prerequisite knowledge, providing learning materials, and offering post-class consultations.

Question 2: Do You Think the Courses You Have Taken Are Useful?

answer：

I believe the entire curriculum is highly effective. The foundational courses equip us with a solid grasp of mathematics, programming, and fundamental computer principles; the specialized courses deepen our understanding of systems, networks, databases, algorithms, and software engineering; while the practical courses enable us to apply this knowledge to real-world system development.

For example, Data Structures and Algorithm Design and Analysis enhance our problem-solving and programming skills; Database Principles and Applications is widely used in various projects; Software Engineering provides insight into the actual software development process; Computer Networks helps us understand how network systems operate and are configured. These courses are beneficial for career development, internships, and graduation projects.

Question 3: How Would You Evaluate the Teaching Quality of Your Instructor?

answer：

From the students 'perspective, I believe the overall teaching quality of the instructors is quite good. Most teachers prepare their lessons diligently, are well-versed in the course content, and effectively organize instruction according to the syllabus, enabling us to clearly understand the course's learning objectives, main content, assessment methods, and its relationship with the professional graduation requirements. For example, in core professional courses, instructors not only impart theoretical knowledge but also incorporate case studies, experiments, or projects to enhance understanding.

助我们理解知识点，像程序设计、数据库、计算机网络、软件工程、人工智能等课程，都比较注重理论与实践结合。

我们评价教师教学质量，不只是凭个人感觉，学校也有比较完整的教学质量评价机制。每学期末，我们都会通过在线评教系统对课程和教师进行评价，评价内容包括教学态度、教学内容、教学方法、课堂组织、作业与考核安排等方面。除了学生评教外，学校和学院还会通过教学督导听课、同行听课、期中教学座谈会、学生信息员反馈、课程目标达成情况分析等方式，对课程教学质量进行综合评价。学校会收集学生和教师反馈，并将学生评教统计结果反馈给学院和教师，用于后续教学改进。

从我们实际感受来看，多数教师比较重视学生反馈。如果学生反映某些内容难度较大、讲课节奏偏快，教师通常会通过课后答疑、增加案例讲解、补充学习资料、调整作业难度或利用在线学习平台发布资源等方式进行改进。对于实践性较强的课程，教师也会通过实验报告、课程项目、阶段检查和答辩等方式了解我们的学习效果。

当然，也有一些可以继续改进的地方。比如个别课程内容较多，授课节奏有时偏快；部分课程如果能增加更多工程案例、课堂互动和项目演示，会更有助于我们理解。因此，我认为教师整体教学质量较好，学校也有较完善的“学生评价—督导检查—课程反馈—教学改进”的质量保障机制，能够持续促进教学质量提升。

问题 4：是否有让你印象不好或需要改进的教师？

答案：
总体上教师都比较负责，没有特别负面的印象。部分学生反馈，一些课程内容难度较高，授课进度有时偏快，学生需要花更多课后时间消化。希望在后续教学中，教师可以增加课堂互动、案例演示和阶段性复习，帮助学生更好地理解重点和难点。

问题 5：哪一门课程最让你印象深刻？为什么？

答案：
不同学生可能答案不同。对我来说，实践类课程和课程设计印象比较深，例如《C程序设计课程设计》《软件工程课程设计》《智能应用系统开发课程设计》和毕业设计。这些课程要求我们完成比较完整的系统或项目，从需求分析、功能设计、编码实现到测试和答辩，能够真正体会工程开发过程。

这类课程让我感受到，学习计算机不仅要理解理论，更要能够把知识转化为可运行、可测试、可展示的系统。

问题 6：项目化教学对你有什么帮助？

答案：
项目化教学最大的好处是让我们不只是“学知识点”，而是学会把知识真正用起

These courses help us grasp key concepts; subjects such as programming, databases, computer networks, software engineering, and artificial intelligence all emphasize the integration of theory and practice.

Our evaluation of teaching quality is not based solely on personal impressions; the university also employs a comprehensive teaching quality assessment mechanism. At the end of each semester, we evaluate courses and instructors through an online teaching evaluation system, covering aspects such as teaching attitude, content, methods, classroom organization, and assignment and assessment arrangements. In addition to student feedback, the university and colleges conduct comprehensive evaluations through teaching supervision observations, peer reviews, mid-term teaching symposiums, feedback from student representatives, and analyses of course objective achievement. The university collects feedback from both students and faculty, and provides the statistical results of student evaluations to the respective colleges and instructors for use in subsequent teaching improvements.

From our practical experience, most teachers place significant importance on student feedback. When students indicate that certain content is difficult or the teaching pace is too fast, teachers typically make improvements by providing after-class Q&A sessions, incorporating more case studies, supplementing learning materials, adjusting assignment difficulty levels, or releasing resources through online learning platforms. For practice-oriented courses, teachers also assess learning outcomes through methods such as lab reports, course projects, periodic assessments, and defense sessions.

Of course, there are still areas for further improvement. For instance, some courses cover extensive content, resulting in a somewhat fast teaching pace; incorporating more engineering case studies, classroom interactions, and project demonstrations in certain courses would enhance our understanding. Therefore, I believe the overall teaching quality is strong, and the university has a well-established quality assurance mechanism—comprising student evaluation, supervisory inspections, course feedback, and teaching improvement—which continuously drives the enhancement of teaching quality.

Question 4: Are There Any Teachers Who Left a Negative Impression on You or Need Improvement?

answer：

Overall, the teachers are quite responsible and leave no particularly negative impressions. Some students reported that certain course content is quite challenging, and the teaching pace is sometimes too fast, requiring students to spend more time reviewing after class. It is hoped that in future instruction, teachers will enhance classroom interaction, incorporate case demonstrations, and conduct periodic reviews to help students better grasp key and difficult concepts.

Question 5: Which Course Impressed You the Most? Why?

answer：

Students may arrive at different conclusions. For me, practical courses and course projects left the deepest impression, such as the "C Programming Course Project," "Software Engineering Course Project," "Intelligent Application System Development Course Project," and the graduation project. These courses required us to complete relatively comprehensive systems or projects, covering everything from requirements analysis and functional design to code implementation, testing, and defense, allowing us to gain a genuine understanding of the engineering development process.

This type of course made me realize that learning computer science requires not only understanding theories but also the ability to transform knowledge into operational, testable, and demonstrable systems.

Question 6: How Has Project-Based Teaching Benefited You?

answer：

The greatest advantage of project-based teaching is that it enables us not merely to ' learn knowledge points,' but to truly apply that knowledge in practice.

来。比如在《C 程序设计》《数据库原理及应用》《软件工程》《智能应用系统开发》等课程中，我们需要围绕一个具体项目完成需求分析、功能设计、代码实现、测试调试、文档撰写和最终汇报。这个过程让我们更清楚地理解课程内容在实际工程中的作用。

项目化教学也明显提升了我们的工程实践能力和问题解决能力。做项目时，经常会遇到课本上没有直接答案的问题，比如程序报错、数据库设计不合理、接口调用失败、系统功能无法正常运行等。我们需要自己查资料、调试代码、和同学讨论，必要时再向老师请教。通过这个过程，我们的问题分析能力、现代工具使用能力和自主学习能力都有明显提升。

另外，项目通常是小组完成的，所以也锻炼了团队协作和沟通能力。我们需要分工、协调进度、整合代码、准备展示和答辩，这和真实的软件开发流程比较接近。通过项目化教学，我们不仅掌握了专业知识，也提前熟悉了工程项目的基本流程，对以后实习、毕业设计和就业都有帮助。

问题 7：每学期是否参与课程评教？你认为能否反映真实教学情况？

答案：
每学期我们都需要参加课程评教，对教师授课情况、课程内容、教学方法、作业安排和学习效果等进行评价。只要学生认真、客观填写，评教基本可以反映课程教学的真实情况。学校也会结合学生评教、督导听课、同行评价和课程考核结果综合判断教学质量。

问题 8：对教务或学院教学管理有什么建议？

答案：
可以建议进一步加强企业合作，提供更多真实项目和实习岗位；优化课程时间安排，避免多门课程的大作业或考试过于集中；增加课程实践内容和案例教学；在高年级课程中多邀请企业工程师开展讲座；为学生提供更多跨方向选修课和技术训练资源。

问题 9：学生期末考试通过率如何？不及格有什么后果和补救措施？

答案：

不同课程通过率会有所差异。学生如果课程不及格，会影响绩点和学分取得。学校提供补考或重修机会，学生可以通过补考或重修完成课程要求。如果课程不及格较多或学分不足，学院会进行学业预警并安排帮扶。学生必须修满培养方案规定学分并达到相关要求，才能毕业和获得学位。

问题 10：学生出勤率如何？迟到或旷课有什么后果？

For example, in courses such as "C Programming," "Database Principles and Applications," "Software Engineering," and "Intelligent Application System Development," we complete a series of tasks centered around a specific project: requirements analysis, functional design, code implementation, testing and debugging, documentation preparation, and final presentation. This process helps us gain a clearer understanding of how the course content applies in real-world engineering scenarios.

Project-based teaching has significantly enhanced our engineering practice skills and problem-solving abilities. When working on projects, we frequently encounter problems without direct solutions in textbooks, such as program errors, poorly designed databases, failed interface calls, or malfunctioning system functions. We must independently research materials, debug code, discuss with classmates, and seek guidance from instructors when necessary. Through this process, our problem analysis skills, proficiency in using modern tools, and self-learning capabilities have all improved markedly.

Moreover, projects are typically completed by teams, thereby enhancing teamwork and communication skills. The process involves task division, schedule coordination, code integration, and preparation for presentations and defenses—all closely mirroring real-world software development workflows. Through project-based learning, we not only acquire professional knowledge but also gain early familiarity with fundamental engineering project procedures, which proves beneficial for future internships, graduation projects, and career development.

Question 7: Do You Participate in Course Evaluation Each Semester? Do You Believe It Accurately Reflects the Actual Teaching Situation?

answer：

Each semester, we participate in course evaluation sessions to assess teachers' instructional performance, course content, teaching methods, assignment design, and learning outcomes. As long as students complete their evaluations diligently and objectively, these assessments accurately reflect the actual teaching quality of the courses. The university also comprehensively evaluates teaching quality by integrating student feedback, supervisory classroom observations, peer reviews, and course assessment results.

Question 8: What Suggestions Do You Have for Academic Affairs or College Teaching Management?

answer：

It is recommended to further strengthen corporate collaboration by providing more authentic projects and internship opportunities; optimize course schedules to avoid excessive concentration of major assignments or exams across multiple courses; enhance practical course content and case-based teaching; invite more industry engineers to deliver lectures in advanced courses; and offer students a wider range of interdisciplinary elective courses and technical training resources.

Question 9: What Is the Pass Rate for Students in the Final Exams? What Are the Consequences of Failing, and What Remedial Measures Are Available?

answer：

Passing rates vary across courses. A course failure affects a student's GPA and credit accumulation. The university offers make-up exams or course retakes, allowing students to meet course requirements. If a student fails multiple courses or falls short of required credits, the college issues an academic alert and provides support. Students must complete all required credits specified in their program and meet all relevant requirements to graduate and receive their degree.

Question 10: What Is the Student Attendance Rate? What Are the Consequences of Being Late or Skipping Class?

答案：

总体出勤率较高。出勤情况是过程性考核的重要组成部分，迟到、早退或旷课会影响平时成绩。如果学生缺课达到学校规定比例，可能会被取消课程考试资格。因此，教师和学院都比较重视课堂出勤管理。

6. 毕业生成果

问题 1：本专业如何帮助你为就业或继续深造做好准备？

答案：
本专业通过课程体系、实践教学、实习实训、竞赛科研和国际交流等多种方式帮助我们为就业或继续深造做准备。

首先，数学、程序设计、数据结构、算法、操作系统、数据库、计算机网络等课程为我们打下理论和技术基础。其次，C 程序设计课程设计、算法课程设计、软件工程课程设计、智能应用系统开发课程设计等实践环节帮助我们把理论知识转化为工程能力。再次，专业见习和专业实习让我们了解企业真实工作流程和岗位要求。第二课堂、学科竞赛和科研训练也提升了我们的创新能力和解决实际问题的能力。

如果选择继续深造，专业课程中的数学基础、算法能力、文献阅读能力、实验分析能力和毕业设计训练，也为研究生阶段学习打下了基础。

7. 国际化，行业合作，学科交叉

问题 1：教学内容如何保持与行业发展同步？

答案：
从学生角度来看，我能明显感受到课程内容并不是固定不变的，而是在不断结合行业发展进行更新。比如，在人工智能、大数据、云计算、网络安全、智能应用系统开发等方向，课程中会逐步加入一些新技术、新工具和新案例。例如在人工智能、机器学习、数据挖掘、智能应用系统开发等课程中，老师会介绍当前比较热门的技术方向和应用场景，让我们了解行业正在关注什么、实际项目中会用到什么。

除了课堂内容，学院也会通过企业讲座、专业见习、专业实习、校企合作项目等方式，让我们接触企业真实需求。比如在专业见习或企业讲座中，我们可以了解到企业对软件开发、数据处理、人工智能应用、网络安全等岗位能力的要求；在专业实习中，也能看到课堂知识和企业实际工作之间的联系。

同时，一些老师本身也参与科研项目、企业实践或行业交流，所以会把新的技术案例、工程经验和实际问题带到课堂或课程项目中。比如在课程设计、毕业设计

answer：

The overall attendance rate is relatively high. Attendance status constitutes a critical component of formative assessment, as tardiness, early departure, or absence from classes may adversely affect regular grades. If a student's absence reaches the threshold specified by the institution, they may be disqualified from taking course examinations. Consequently, both faculty members and the college place significant emphasis on classroom attendance management.

6. Graduation Achievements

Question 1: How Does This Program Prepare You for Employment or Further Academic Pursuits?

answer：

This program equips students with comprehensive preparation for employment or further studies through a diversified approach encompassing curriculum design, practical instruction, internships, competition participation, research opportunities, and international exchanges.

First, courses such as mathematics, programming, data structures, algorithms, operating systems, databases, and computer networks lay the theoretical and technical foundation for us. Second, practical components—including C programming course projects, algorithm course projects, software engineering course projects, and intelligent application system development projects—help us transform theoretical knowledge into engineering skills. Third, professional internships and field placements enable us to understand real-world corporate workflows and job requirements. Additionally, extracurricular activities, academic competitions, and research training enhance our innovation capabilities and problem-solving skills.

If one chooses to pursue further studies, the foundational mathematics skills, algorithmic proficiency, literature analysis capabilities, experimental analytical abilities, and graduation project training covered in the specialized curriculum also lay a solid foundation for graduate-level learning.

7. Internationalization, Industry Collaboration, and Interdisciplinary Integration

Question 1: How Can Teaching Content Stay Aligned With Industry Developments?

answer：

From the student's perspective, I can clearly perceive that the course content is not static but is continuously updated in alignment with industry developments. For instance, in areas such as artificial intelligence, big data, cloud computing, cybersecurity, and intelligent application system development, the curriculum progressively incorporates new technologies, tools, and case studies. In courses covering artificial intelligence, machine learning, data mining, and intelligent application system development, instructors introduce currently prominent technical trends and application scenarios, enabling us to understand what the industry is focusing on and what technologies are actually employed in real-world projects.

Beyond classroom instruction, the college also exposes students to real-world industry needs through corporate lectures, professional internships, practical training programs, and school-enterprise collaboration projects. For instance, during internships or industry talks, students gain insights into the skill requirements for roles in software development, data processing, artificial intelligence applications, and cybersecurity; internship experiences further demonstrate the connection between theoretical knowledge and actual workplace practices.

Meanwhile, some teachers themselves participate in research projects, corporate internships, or industry exchanges, enabling them to incorporate novel technological cases, engineering experiences, and practical challenges into their classrooms or course projects—for instance, in curriculum design and graduation projects.

和第二课堂竞赛中，选题有时会结合智能应用、大数据分析、Web 系统开发等实际场景，这让我们觉得学习内容更贴近行业需求。

另外，我们也可以通过课程评教、座谈会、workload 调查等方式反馈课程内容是否实用、是否需要增加实践或新技术内容。学院会根据学生反馈、毕业生反馈和企业反馈持续调整课程体系。因此，我认为本专业的教学内容能够通过“课程更新—企业参与—教师实践—学生反馈”的方式保持与行业发展同步。

问题 2：是否有跨专业或团队合作项目？

答案：
有。从学生角度来看，本专业不仅有很多课程项目采用小组合作形式完成，也有不少跨专业合作的机会。课程项目中，每个小组通常会进行明确分工，例如需求分析、系统设计、编码实现、测试调试、文档撰写和成果汇报。虽然这些项目多数是在本专业内部完成，但项目内容往往涉及具体应用场景、用户需求分析、数据处理、界面设计和系统部署，因此本身就具有一定的跨领域特点。

除了课程项目，在第二课堂、学科竞赛和大学生创新创业训练项目中，我们有更多机会和其他专业的同学合作。例如，在数学建模竞赛或相关项目中，计算机专业学生可以和数学专业学生组队，数学专业同学更擅长模型构建、理论推导和数据分析，我们则更多负责算法实现、程序设计、数据处理和结果可视化。这样的合作可以把数学建模能力和计算机实现能力结合起来，完成更加完整的建模与求解过程。

另外，在大学生创新创业项目中，也会有和医药、化工类专业同学合作的机会。比如围绕智能医疗、健康数据分析、化工生产数据监测、实验数据管理等主题，医药或化工专业同学更了解专业背景、业务流程和实际需求，计算机专业同学则负责系统设计、数据库建设、数据分析、Web 或移动端开发等技术实现。通过这样的跨专业合作，我们能够更好地理解真实应用场景，也能学习如何把计算机技术应用到其他行业中。

因此，我认为跨专业和团队合作项目对我们帮助很大。一方面，它提升了我们的团队协作、沟通表达和项目管理能力；另一方面，也让我们认识到计算机专业不是孤立的，而是可以和数学、医药、化工、管理等多个领域结合，解决更加实际和复杂的问题。

问题 3：学院有没有支持国外交流学习的项目？

答案：
有。学校和学院提供一定的国际交流和短期学习机会，例如赴澳门地区高校、澳大利亚高校等进行短期交流或学习。学院也会通过国际合作项目、学术交流、国际课程或海外学习信息发布等方式，帮助学生拓展国际视野。

学生可以通过学院通知、国际合作部门网站、辅导员或负责国际化工作的教师了解相关项目。

In the second-classroom competition, topics are sometimes integrated with real-world scenarios such as intelligent applications, big data analytics, and web system development, making the learning content more aligned with industry needs.

Additionally, we can gather feedback on the practicality of course content and the need for incorporating more practical or cutting-edge technological elements through methods such as course evaluations, seminars, and workload surveys. The college continuously refines the curriculum system based on student feedback, graduate feedback, and industry insights. Therefore, I believe the teaching content of this program remains aligned with industry developments through a cyclical process of "course updates—industry participation—teacher practice—student feedback."

Question 2: Are There Any Interdisciplinary or Team-Based Projects?

answer：

Yes. From the students' perspective, this major not only features numerous course projects completed through group collaboration but also offers ample opportunities for interdisciplinary cooperation. Within these projects, each group typically assumes clearly defined roles, such as requirement analysis, system design, coding implementation, testing and debugging, documentation preparation, and project presentation. Although most projects are completed within the major itself, their content often involves specific application scenarios, user requirement analysis, data processing, interface design, and system deployment, thereby inherently possessing interdisciplinary characteristics.

Beyond course projects, we have ample opportunities to collaborate with students from other disciplines in extracurricular activities, academic competitions, and university innovation and entrepreneurship training programs. For instance, in mathematical modeling competitions or related initiatives, computer science students can team up with mathematics majors. Mathematics students excel in model construction, theoretical derivation, and data analysis, while we focus on algorithm implementation, programming, data processing, and result visualization. Such collaboration integrates mathematical modeling capabilities with computational implementation skills, enabling the completion of a more comprehensive modeling and solution process.

Furthermore, in university student innovation and entrepreneurship projects, there are opportunities to collaborate with peers from pharmaceutical and chemical engineering disciplines. For instance, in initiatives focused on intelligent healthcare, health data analysis, chemical production data monitoring, and experimental data management, students from these fields possess deeper expertise in their professional backgrounds, business processes, and practical requirements, while computer science students handle technical implementation tasks such as system design, database development, data analysis, and web or mobile application development. Through such interdisciplinary collaboration, we gain a better understanding of real-world application scenarios and learn how to apply computer technologies across various industries.

Therefore, I believe interdisciplinary and teamwork projects have been immensely beneficial to us. On one hand, they enhanced our teamwork, communication, and project management skills; on the other hand, they made us realize that computer science is not an isolated discipline but can be integrated with fields such as mathematics, medicine, chemical engineering, and management to address more practical and complex challenges.

Question 3: Does the College Offer Any Programs Supporting International Exchange and Study?

answer：

Yes. Schools and colleges offer various international exchange and short-term study opportunities, such as programs at universities in Macau or Australia. Additionally, the college helps students broaden their global perspectives through international cooperation projects, academic exchanges, international curricula, and the dissemination of overseas study information.

Students can learn about relevant programs through college notifications, the International Cooperation Department website, counselors, or faculty members responsible for international affairs.

问题 4：学院为海外学习提供哪些支持？

答案：
学院会提供项目宣传、申请指导、语言能力提升、跨文化适应指导、课程匹配建议、手续办理协助等支持。有些项目还可以申请学校或学院层面的奖学金资助。学生返校后，学院也会根据学校规定协助进行学分认定或交流成果认定。

这些措施帮助学生降低出国交流的难度，并提升跨文化交流能力。

问题 5：你从哪些渠道了解出国交流项目？

答案：
主要通过学校国际合作部门网站、学院通知、辅导员通知、班级群、负责国际化工作的教师以及学长学姐分享等渠道了解相关信息。

问题 6：你是否有去德国或其他国家学习的计划？

答案：
可以根据个人情况回答。如果有计划，可以说明目标国家、专业方向和原因，例如希望通过海外学习拓展国际视野、提升专业能力、了解不同教育体系。如果暂时没有计划，也可以说明目前主要希望完成本科课程学习和专业实践，未来如果有合适机会，会考虑参加短期交流或继续深造。

问题 7：学校多久举办学术讲座？你们是否参加？

答案：
学院会不定期举办学术讲座、企业讲座和技术分享，内容包括人工智能、大数据、网络安全、软件开发、职业发展等。学生可以根据兴趣参加。这些讲座有助于我们了解学科前沿、行业趋势和职业发展方向。

Question 4: What Support Does the College Offer for Overseas Studies?

answer：

The college provides comprehensive support including project promotion, application guidance, language proficiency enhancement, cross-cultural adaptation counseling, course matching recommendations, and assistance with administrative procedures. Some programs also offer scholarship opportunities at the university or college level. Upon returning to campus, the college will facilitate credit recognition or evaluation of exchange achievements in accordance with university regulations.

These measures help students reduce the difficulties of studying abroad and enhance their intercultural communication skills.

Question 5: Through Which Channels Did You Learn About the Overseas Exchange Programs?

answer：

The relevant information is primarily obtained through channels such as the university's International Cooperation Department website, college announcements, counselor notifications, class groups, faculty members responsible for international affairs, and sharing by senior students.

Question 6: Do You Have Any Plans to Study in Germany or Other Countries?

answer：

The response can be tailored to individual circumstances. If you have specific plans, you may outline your target country, academic focus, and motivations—for instance, aiming to broaden your international perspective, enhance professional skills, or gain insights into different education systems through overseas study. If you currently have no immediate plans, you could state that your primary focus is completing your undergraduate degree and gaining professional experience, with the intention to consider short-term exchange programs or further studies if suitable opportunities arise in the future.

Question 7: How Often Does the School Hold Academic Lectures? Do You Attend Them?

answer：

The academy periodically hosts academic lectures, corporate seminars, and technical sharing sessions covering topics such as artificial intelligence, big data, cybersecurity, software development, and career development. Students may attend based on their interests. These lectures provide insights into the latest advancements in academic disciplines, industry trends, and career pathways.

计算机科学与技术专业人才培养方案

专业代码： 080901 **专业类：** 计算机类 **授予学位：** 工学学士

一、培养目标

本专业面向适应区域经济与社会发展需求，培养具有坚定的社会主义理想信念、德智体美劳全面发展，具备科学和人文素养、职业道德和国际视野，具有创新意识和团队精神，具备系统分析、开发、实践能力和项目组织管理能力，能够应用专业基础理论和技术解决计算机应用领域中的复杂工程问题，适应技术进步和社会需求变化，成为能够在 IT 行业、科研机构、企事业单位从事计算机应用领域的项目设计、开发、管理、服务等方面工作的应用型人才。

本专业学生毕业 5 年左右，要求能达到以下预期目标:

1. 具有扎实的数理、计算机专业基础理论知识和专业技能，具有考虑环境、安全、社会可持续发展等工程与环境因素，解决计算机应用领域复杂工程问题;

2. 具备较强的工程实践能力、团队协作和工程管理能力，并具有良好的沟通交流能力。能够成为在 IT 行业、科研机构、企事业单位中从事计算机应用领域的系统分析、设计、开发和测试的技术骨干或项目管理人员;

3. 具有良好的科学文化素养和思想道德素养，身心健康，具备优良的职业道德和社会责任感，掌握职业相关的经济、管理和法律等知识;

4. 具有创新意识和终身学习能力，能够通过多种学习渠道不

断更新和调整自身的核心知识和能力，具有一定的国际视野和跨文化交流、竞争与合作能力，适应技术进步和社会发展。

二、毕业要求

学生在毕业时应达到以下具体要求:

1. 工程知识: 能够将数学、自然科学、计算、工程基础和计算机相关专业理论知识解决计算机应用领域中的复杂工程问题。

1.1 具备解决复杂工程问题所需的自然科学知识、数学知识、计算科学，选择恰当的数学模型描述计算机应用系统，并对模型具备一定的分析、比较和评价能力;

1.2 具备计算机应用系统开发所需的相关工程基础知识，能够运用计算机相关专业的基本理论，解决计算机应用系统设计与开发涉及的相关复杂工程问题。

2. 问题分析: 能够应用数学、自然科学和工程科学的基本原理，识别、表达、并通过文献研究分析计算机应用系统领域中的复杂工程问题，综合考虑可持续发展的要求，同时能对其合理性进行评价并获得有效结论。

2.1 能够运用数学、自然科学、工程基础和计算机专业理论和技术方法识别、表达计算机应用领域的复杂工程问题;

2.2 针对复杂的计算机应用系统、过程或问题，能通过研究分析相关文献寻求解决方案，综合考虑可持续发展的要求，达到应用要求或加以改进。

3. 设计/开发解决方案: 能够设计针对计算机应用系统中的复杂工程问题的具体方案，如智能应用系统、大数据分析系统、移动应用系统等系统设计，并能够在设计环节中体现创新性，并

从健康与安全、全生命周期成本与净零碳要求、法律与伦理、社会与文化等角度考虑可行性。

3.1　能根据智能应用系统、大数据分析系统、移动应用系统等系统需求或设计目标确定具体的解决方案，并从健康与安全、全生命周期成本与净零碳要求、法律与伦理、社会与文化等角度考虑可行性;

3.2　能够在设计环节中体现创新性，创新的解决复杂问题。

4. 研究：能够基于科学的理论与方法，对计算机应用系统中的复杂工程问题进行研究，选择研究方案，设计实验，对实验数据进行分析，并通过信息综合得到合理有效的结论。

4.1　能针对计算机应用系统中的复杂工程问题，基于计算机科学理论与方法，选择研究路线，设计可行的实验方案;

4.2　能针对计算机应用系统中的复杂工程问题，结合专业理论与实践对实验结果进行科学的分析和解释，并通过信息综合得出合理有效的结论。

5. 使用现代工具：针对工程问题，在计算机软硬件设计与开发、系统分析设计等过程中，选择和使用恰当的现代工程开发工具和信息技术开发工具，解决计算机应用领域的工程实践中的复杂问题，并能够理解其局限性。

5.1　掌握计算机专业主要资料文献获取方法，能够利用能够运用现代信息技术进行文献检索和资料查询;

5.2　能够运用常见建模工具、设计工具、开发工具、测试工具，解决计算机应用系统复杂工程问题，并理解其工作原理及其局限性。

6. 工程与可持续发展：在解决计算机应用领域复杂工程问题时，能够基于工程相关背景知识，分析和评价工程实践对健康、安全、环境、法律以及经济和社会可持续发展的影响，并理解应承担的责任。

6.1 熟悉 IT 行业相关的技术标准、知识产权、产业政策和法律法规;

6.2 具有工程实践的经历，了解计算机工程相关的背景知识，能够就计算机工程实践和应用领域复杂工程问题解决方案对健康、安全、环境、法律以及经济和社会可持续发展的影响有正确认识，并理解应承担的责任。

7. 工程伦理和职业规范：有工程报国、为民造福的意识，具有人文社会科学素养和社会责任感，能够理解和践行工程伦理，在计算机工程实践中遵守工程职业道德、规范和相关法律，履行责任。

7.1 具备正确的世界观、人生观和价值观，能够不断地提高自身的人文社会科学素养，身心健康;

7.2 恪守工程伦理，能够在计算机工程实践中遵守专业工程师职业道德、规范和相关法律，履行社会责任。

8. 个人与团队：具有团队协作精神，能够在多样化、多学科背景下的团队中承担个体、团队成员以及负责人的角色。

8.1 能够在多样化、多学科背景下理解团队中每个角色的定位与责任，能胜任个人承担的角色任务;

8.2 能够在多样化、多学科背景下与团队其他成员有效沟通，听取并综合团队其他成员的意见与建议，做出合理决策。

9. 沟通：能够就计算机应用领域复杂工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和、陈述发言、清晰表达或回应指令；能够在跨文化背景下进行沟通和交流，理解、尊重语言和文化差异。

9.1 具有阅读和撰写技术文档、陈述发言、清晰表达或回应指令的能力与业界同行及客户进行有效沟通和交流，系统、有条理地表达计算机应用领域复杂工程问题；

9.2 能理解和尊重不同文化的差异性和多样性，具备跨文化交流的语言和书面表达能力，能就计算机科学与技术相关领域专业问题，在跨文化背景下进行沟通和交流。

10. 项目管理：能够将与工程项目相关的管理原理和经济决策的方法用于多学科交叉的计算机工程领域复杂工程问题。

10.1 掌握与计算机工程领域项目相关的管理和经济决策的知识、原理和方法；

10.2 能够将工程管理的原理和经济决策的方法用于在多学科交叉环境下的应用系统的设计、运行及管理。

11. 终身学习：具有自主学习、终身学习和批判性思维的意识和能力，能理解广泛的技术变革对工程和社会的影响，适应新一代信息技术变革。

11.1 能认识到计算机技术的迅速发展对社会的影响，意识到自我探索和终身学习的必要性，具备自主学习能力；

11.2 解计算机科学与技术学科相关技术与理论的重要进展和前沿动态。理解技术变革对工程和社会的影响，适应新一代信息技术变革。

三、修业年限

基本学制 4 年，修业年限 3-6 年。

四、主干学科

计算机科学与技术。

五、核心课程

C 程序设计、离散数学、计算机组成原理、数据结构、数据库原理及应用、算法设计与分析、操作系统、计算机网络、软件工程、人工智能。

六、主要教学实践环节

军事训练、专业见习、C 程序设计课程设计、算法分析与设计课程设计、软件工程课程设计、智能应用系统开发课程设计、专业实习、毕业设计（论文）、第二课堂等。

七、学分、学时分配

课程性质	课程类别	学时数	比例（%）	学分数	比例（%）
通识教育课程	必修	704	32.1	38	24.5
	选修	80	3.6	5	3.2
学科基础课程	必修	864	39.4	54	34.8
专业课程	必修	336	15.3	21	13.5
	选修	208	9.6	13	8.5
集中实践教学环节	必修	/	/	24	15.5
合计		2192	100	155	100

八、毕业条件与学位授予

1. 毕业条件

在规定的修业年限内修完培养方案规定的全部课程，成绩合格。修满规定的最低总学分 155 学分，其中必修课 137 学分（含集中实践教学环节 24 学分），选修课最低 18 学分。

2. 学位授予

符合学校学士学位授予条件的，授予工学学士学位。

九、教学时间分配表

学 期 项目及周数	第一学年		第二学年		第三学年		第四学年		合 计
	第一学期	第二学期	第三学期	第四学期	第五学期	第六学期	第七学期	第八学期	
报到、入学教育和国防教育	2								2
课堂教学	16	16	16	16	16	16			96
复习考试	1	1.5	1.5	1.5	1.5	1.5			8.5
专业见习	*	*	*	*	*	*	*		2
专业实习							6		6
毕业设计（论文）							5	11	16
劳动教育	*	*	*	*	*	*	*	*	
第二课堂	*	*	*	*	*	*	*	*	
机 动		2.5	2.5	2.5	2.5	2.5	7	7	16.5
寒暑假	12		12		12		3.5		39.5
教育周数	19	20	20	20	20	20	18	18	155
学年周数	51		52		52		39.5		194.5

十、教学安排表

（一）通识教育课程

课程类别	课程号	课程名称	学时		学分	开课学期、教学周数与周学时分配								考核形式		备注
			讲授	实践		1	2	3	4	5	6	7	8	考试	考查	
必修	00001115	思想道德与法治	32	16	3	2+1								√		
	00001112	中国近现代史纲要	48		3		3							√		
	00001116	马克思主义基本原理	48		3			3						√		
	00001117	毛泽东思想和中国特色社会主义理论体系概论	32	16	3				2+1					√		
	00001118	习近平新时代中国特色社会主义思想概论	48		3					3				√		
	00001105	形势与政策	64		2	*	*	*	*	*	*	*	*		√	*每学期 8 学时
	00001129	大学英语	128		8	4	4							√		
	00001007	大学体育	16	112	4	2*	2*	2*	2*					√		*实践各 4 学时
	00001020	军事理论	32		2		2								√	
	00001063	职业规划	16		1		1								√	
	00001061	大学生创新基础	16		1				1*						√	
	00001062	就业指导与创业基础	16		1					1					√	
	00001131	大学生劳动教育	16		1	1	*	*	*	*	*	*			√	*另安排课外实践 48 学时
	00001132	国家安全教育专题	16		1	*	*	*	*	*	*	*			√	
	00002013	大学生心理健康教育	32		2	*	*	*	*	*	*	*			√	*另安排课外实践 16 学时在第 2-7 学期完成
	小计		**704**		**38**	**10***	**12***	**5***	**6***	**4***	*	*				
选修		人文与科学素养选修课程 人文与社会科学类	32		2		*	*	*	*	*	*	*		√	至少选修 2 学分
		创新创业类	32		2		*	*	*	*	*	*	*		√	
		中华民族共同体概论	32		2		*	*	*	*					√	
		公共艺术课程 美学和艺术史论类	16		1		*	*							√	至少选修 2 学分，其中美学和艺术史论类、艺术鉴赏和评论类至少 1 学分。音乐和美术互选。
		艺术鉴赏和评论类	16		1		*	*							√	
		艺术体验和实践类	16		1		*	*	*	*	*	*	*		√	
		“四史”教育类课程		16		1		*	*	*	*					√
	小计		**80**		**5**											

（二）学科基础课程

课程类别	课程号	课程名称	学时		学分	开课学期、教学周数与周学时分配								考核形式		备注
			讲授	实践		1	2	3	4	5	6	7	8	考试	考查	
必修	00001071A	高等数学 A	160		10	5	5							√		
	00001072A	线性代数 A	36	12	3	3								√		
	24001002	Python 程序设计	32	16	3	3								√		
	00001019B	大学物理 B	48	16	4		4							√		
	24001012	C 程序设计	48	16	4		4							√		
	24011011	离散数学	48		3		3							√		
	24001005	数字逻辑电路	32	16	3			3						√		
	00001074A	概率论与数理统计 A	40	8	3			3						√		
	24001006	数据结构	48	32	5			5						√		
	24001007	数据库原理及应用	48	16	4			4						√		
	24001008	计算机网络	48	16	4				4					√		
	24001013	计算机组成原理	48	16	4				4					√		
	24001010	操作系统	48		3				3					√		
	24001011	工程伦理	16		1						1				√	
	小计		**864**		**54**	**11**	**16**	**15**	**11**		**1**					

（三）专业课程

课程类别		课程号	课程名称	学时		学分	开课学期、教学周数与周学时分配								考核形式		备注
				讲授	实践		1	2	3	4	5	6	7	8	考试	考查	
必修		24012034	计算机科学导论	16		1	1									√	
		24012035	算法设计与分析	32	16	3				3					√		
		24012036	Java 程序设计	32	16	3				3					√		
		24014065	人工智能	32	16	3					3					√	
		24012003	编译原理	32		2					2				√		
		24012040	数据挖掘基础	32	16	3					3				√		
		24012041	软件工程	32	16	3						3			√		
		24012042	智能应用系统开发	32	16	3					3					√	
		小计		**336**		**21**	**1**	**0**	**0**	**6**	**11**	**3**					
选修	限选	课群一：生物医学信息															
		24014043	Java 程序设计进阶	32	16	3				*	*	*	*			√	至少选修 3 学分
		24014044	移动应用程序开发	32	16	3				*	*	*	*			√	

		24013045	软件项目管理	32	16	3				*	*	*	*			√	
		24013046	人机交互技术	32	16	3				*	*	*	*			√	
		24013057	医疗大数据	32	16	3				*	*	*	*			√	
		24013058	医学影像处理	32	16	3				*	*	*	*			√	
		24013059	计算生物学	32	16	3				*	*	*	*			√	
		小计		**48**		**3**											
		课群二：智能设计与制造															
		24013055	机器学习基础	32	16	3				*	*	*	*			√	至少选修3学分
		24013047	数字图像处理	32	16	3				*	*	*	*			√	
		24042006	数据可视化技术	32	16	3				*	*	*	*			√	
		24013048	机器人技术与应用	32	16	3				*	*	*	*			√	
		24013060	工业互联网	32	16	3				*	*	*	*			√	
		24013061	数字孪生技术	32	16	3				*	*	*	*			√	
		24013062	智能控制原理	32	16	3				*	*	*	*			√	
		小计		**48**		**3**											
	任选	24013028	计算机英语	32		2						2				√	至少选修7学分
		24012038	Linux 操作系统	16	16	2					2					√	
		24014045	云计算基础	16	16	2						*	*			√	
		24014046	智慧医疗系统	16	16	2						*	*			√	
		24014047	Web 应用开发	32	16	3						3				√	
		24014048	程序设计竞赛创新训练	16	16	2				*	*	*	*			√	
		24014049	网络安全	32	16	3					3					√	
		24014050	科技论文写作	16		1							1			√	
		24014066	Hadoop 基础	16	16	2						*	*			√	
		24014067	Spark 基础	16	16	2						*	*			√	
		24014059	大数据开发技术	16	16	2						*	*			√	
		24014060	AI 大模型应用开发	16	16	2						*	*			√	
		小计		**112**		**7**					**5***	**5***	1*				

（四）集中实践教学环节

编号	环节名称	学分	周数	形式		各学期周数分配							
				集中	分散	第一学期	第二学期	第三学期	第四学期	第五学期	第六学期	第七学期	第八学期
00001702	军事训练（含入学教育）	2	2	√		2							
24006001	专业见习	1	2	√		*	*	*	*	*	*	*	
24017002	C 程序设计课程设计	1	1	√			1						
24016017	算法设计与分析课程设计	2	2	√						2			
24016018	软件工程课程设计	2	2	√							2		
24016019	智能应用系统开发课程设计	1	1	√							1		
24006002	专业实习	3	6	√								6	

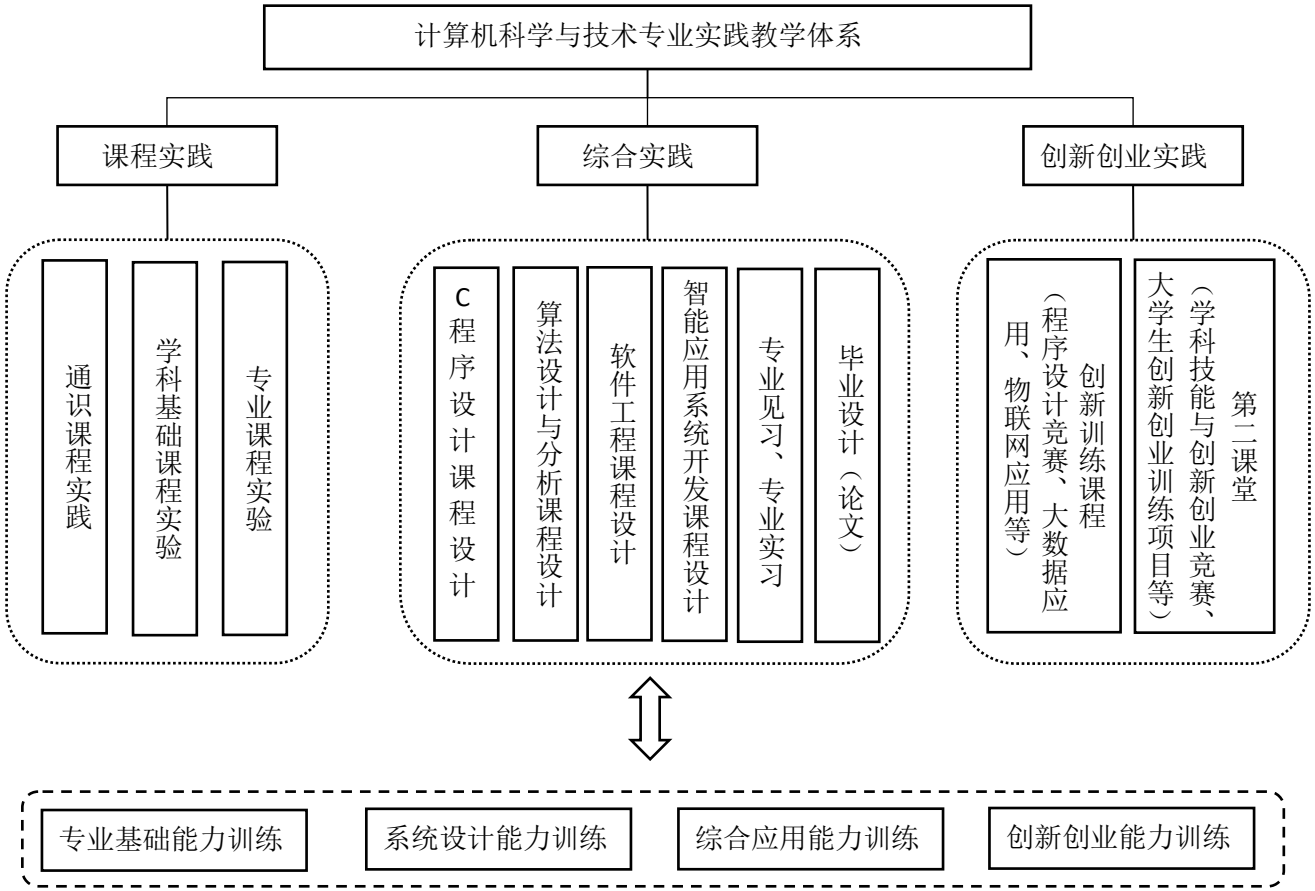
24006003	毕业设计（论文）	8	16	√								5	11
00015300	第二课堂素质拓展	4	16		√	*	*	*	*	*	*	*	*
小计		**24**	**48**			**2**	**1**	**1**	**1**	**2**	**3**	**11**	**11**

注：第二课堂素质拓展学分认定按学校相关规定执行。

十一、专业实践教学学分统计表

课程性质	通识教育课程	学科基础课程	专业课程		集中实践教学环节	合计
			必修	选修		
实践教学学分数	5	10.3	6	3	24	51.3
占总学分比例（%）	3.2	6.6	3.9	1.9	15.5	31.1
实践教学学时数	144	164	96	48	/	452
占总学时比例（%）	6.6	7.5	4.4	2.2	/	20.7

十二、专业实践教学体系逻辑图



十三、毕业要求对培养目标支撑矩阵

毕业要求	培养目标 1	培养目标 2	培养目标 3	培养目标 4
1	√			
2	√			
3	√	√		
4	√	√		
5		√		
6	√	√	√	
7			√	
8		√		
9		√		√
10		√		
11				√

十四、课程对毕业要求支撑矩阵

课程名称	毕业要求 1		毕业要求 2		毕业要求 3		毕业要求 4		毕业要求 5		毕业要求 6		毕业要求 7		毕业要求 8		毕业要求 9		毕业要求 10		毕业要求 11	
	1.1	1.2	2.1	2.2	3.1	3.2	4.1	4.2	5.1	5.2	6.1	6.2	7.1	7.2	8.1	8.2	9.1	9.2	10.1	10.2	11.1	11.2
思想道德与法治					M							H		M								
中国近现代史纲要					L									H								
马克思主义基本原理					L								M	H								
毛泽东思想和中国特色社会主义理论体系概论					L								M	H								
习近平新时代中国特色社会主义思想概论					L								M	H								
形势与政策					M						H		L									
大学英语				L													H	M				L
大学体育													H		L	L						
军事理论												L	M	H	L	L						
职业规划														L							H	
大学生创新基础						H		M														
就业指导与创业基础															L	L			M	M	H	
大学生劳动教育													H	M	L	L						
国家安全教育专题					M							H		M								
大学生心理健康教育													H									
“四史”教育类													H	M								
高等数学	H	M	H				M	M														
线性代数	H	M	H				M	M														
Python 程序设计			M					L		H												

课程名称	毕业要求 1		毕业要求 2		毕业要求 3		毕业要求 4		毕业要求 5		毕业要求 6		毕业要求 7		毕业要求 8		毕业要求 9		毕业要求 10		毕业要求 11	
	1.1	1.2	2.1	2.2	3.1	3.2	4.1	4.2	5.1	5.2	6.1	6.2	7.1	7.2	8.1	8.2	9.1	9.2	10.1	10.2	11.1	11.2
大学物理 B	H		M				H	L														
C 程序设计			M					L		H												
离散数学	H	M	H				M	M														
数字逻辑电路		L	H							M												
概率论与数理统计	H	M	H				M	M														
数据结构			H		H		M			M												
数据库原理及应用			H		M					H												
计算机网络		H	H		M					M												
计算机组成原理		H	H		H		M	M														
操作系统		H	H		M																	
工程伦理					L						L	H		H								
计算机科学导论		M	H		L				H													H
计算机英语																	H	M				
Linux 操作系统					M				M													
算法设计与分析	M		H		M	M	H															
Java 程序设计			M					L		H												
人工智能		H	H		M																	
编译原理		H	M							L												
数据挖掘基础		M	H		M																	
软件工程					H					M									H			
智能应用系统开发			M	L	H																	
军事训练（含入学教育）													H	M	L	L						

课程名称	毕业要求 1		毕业要求 2		毕业要求 3		毕业要求 4		毕业要求 5		毕业要求 6		毕业要求 7		毕业要求 8		毕业要求 9		毕业要求 10		毕业要求 11	
	1.1	1.2	2.1	2.2	3.1	3.2	4.1	4.2	5.1	5.2	6.1	6.2	7.1	7.2	8.1	8.2	9.1	9.2	10.1	10.2	11.1	11.2
专业见习											L	M		H								L
C 程序设计课程设计							M	M												L		
算法设计与分析课程设计			M				M	H											L			
软件工程课程设计				H	M					M					M	M				H		
智能应用系统开发课程设计					M	H				M					M	M						
专业实习					M								M	M	H	H					L	
毕业设计（论文）					M	M			L			M					H	L				
第二课堂素质拓展							M	M		M		H	H	M								

注：H 表示强支撑，M 表示中等支撑，L 表示弱支撑

Taizhou University

Self-Assessment Report for ASIIN Programme Accreditation for the Bachelor Degree of Computer Science and Technology

Appendix A

Cultivation Plan of Computer Science and Technology

Department of Computer Science and Technology
School of Information Engineering

2025

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Cultivation Plan of Computer Science and Technology

Formulated by: Lu Zenan, Wang Yang

Reviewer: Ren Xiangmin

I . Training Objectives

This programme is oriented toward meeting the needs of regional economic and social development. It aims to cultivate application-oriented talents who possess firm socialist ideals and beliefs, develop comprehensively in morality, intelligence, physical fitness, aesthetics, and labor, and are equipped with scientific and humanistic literacy, professional ethics, and an international perspective.

Students are expected to have innovative awareness and teamwork spirit, as well as capabilities in system analysis, development, practice, and project organization and management. They should be able to apply fundamental theories and technologies to solve complex engineering problems in the field of computer applications, adapt to technological progress and changes in social needs, and become application-oriented professionals engaged in project design, development, management, and service in IT industries, research institutions, enterprises, and public organizations.

Graduates of this programme are expected to achieve the following objectives about five years after graduation:

- Possess solid theoretical knowledge of mathematics and computer science, as well as professional skills, and be able to solve complex engineering problems in the field of computer applications while considering environmental, safety, and sustainable development factors.
- Possess strong engineering practice ability, teamwork ability, and engineering management ability, as well as good communication skills. They can become technical backbones or project managers engaged in system analysis, design, development, and testing in IT industries, research institutions, enterprises, and public organizations.

- Possess good scientific and cultural literacy as well as moral qualities, be physically and mentally healthy, have good professional ethics and social responsibility, and master knowledge related to economics, management, and law.

- Possess innovative awareness and lifelong learning ability, be able to continuously update and adjust their core knowledge and abilities through multiple learning channels, have an international perspective and cross-cultural communication, competition, and cooperation abilities, and adapt to technological progress and social development.

II. Graduation Requirements

Students should meet the following requirements upon graduation:

(1) Engineering Knowledge

Be able to apply knowledge of mathematics, natural sciences, computing, engineering fundamentals, and computer science to solve complex engineering problems in the field of computer applications.

- Possess knowledge of natural sciences, mathematics, and computational science required for solving complex engineering problems; be able to select appropriate mathematical models to describe computer application systems and have the ability to analyze, compare, and evaluate these models.

- Possess relevant engineering fundamental knowledge required for the development of computer application systems; be able to apply basic theories of computer science to solve complex engineering problems involved in system design and development.

(2) Problem Analysis

Be able to apply basic principles of mathematics, natural sciences, and engineering sciences to identify, formulate, and analyze complex engineering problems in the field of computer application systems through literature research, while comprehensively considering the requirements of sustainable development, and be able to evaluate their rationality and obtain effective conclusions.

- Be able to identify and formulate complex engineering problems using mathematics, natural sciences, engineering fundamentals, and computer science theories and technical

methods.

- Be able to analyze complex computer application systems, processes, or problems through literature research to seek solutions, while comprehensively considering the requirements of sustainable development, so as to meet application needs or achieve improvement.

(3) Design/Development of Solutions

Be able to design specific solutions for complex engineering problems in computer application systems, such as intelligent application systems, big data analysis systems, and mobile application systems, and demonstrate innovation in the design process. The feasibility of solutions should be considered from the perspectives of health and safety, life-cycle cost and net-zero carbon requirements, as well as legal, ethical, social, and cultural factors.

- Be able to determine specific solutions based on system requirements or design objectives and consider feasibility from multiple perspectives including safety, sustainability, law, and society.
- Be able to demonstrate innovation in the design process and solve complex problems in an innovative manner.

(4) Research

Be able to conduct research on complex engineering problems in computer application systems based on scientific theories and methods, including selecting research approaches, designing experiments, analyzing experimental data, and obtaining reasonable and effective conclusions through information synthesis.

- Be able to select research approaches and design feasible experimental schemes.
- Be able to analyze and interpret experimental results scientifically and draw reasonable conclusions.

(5) Use of Modern Tools

For engineering problems, be able to select and use appropriate modern engineering development tools and information technology tools in the processes of computer hardware and software design, development, and system analysis, to solve complex engineering problems in computer application practice, and understand their limitations.

- Be able to master methods for acquiring professional literature and information and use modern information technologies for literature retrieval and information search.
- Be able to use modeling, design, development, and testing tools to solve complex problems and understand their working principles and limitations.

(6) Engineering and Sustainable Development

When solving complex engineering problems in computer applications, be able to analyze and evaluate the impact of engineering practice on health, safety, environment, law, and economic and social sustainable development based on engineering background knowledge, and understand corresponding responsibilities.

- Be familiar with technical standards, intellectual property, industrial policies, and laws and regulations in the IT industry.
- Be able to evaluate the impacts of engineering solutions and understand responsibilities.

(7) Engineering Ethics and Professional Conduct

Possess awareness of serving the country through engineering and benefiting society, have humanistic and social science literacy and social responsibility, understand and practice engineering ethics, comply with professional norms and laws, and fulfill responsibilities.

- Possess correct values and continuously improve humanistic literacy, with sound physical and mental health.
- Adhere to engineering ethics and fulfill social responsibilities.

(8)Individual and Teamwork

Possess teamwork spirit and be able to undertake roles as an individual, team member, and leader in multidisciplinary teams.

- Be able to understand roles and responsibilities in teams.
- Be able to communicate and collaborate effectively and make decisions.

(9) Communication

Be able to effectively communicate with peers and the public on complex engineering problems, including writing reports, delivering presentations, clearly expressing ideas, and responding to instructions, and be able to communicate in cross-cultural contexts.

- Be able to express complex problems clearly and systematically.
- Be able to communicate across cultures.

(10) Project Management

Be able to apply management principles and economic decision-making methods related to engineering projects in multidisciplinary environments.

- Master relevant knowledge and methods.
- Apply them in system design, operation, and management.

(11) Lifelong Learning

Possess awareness and ability of self-directed learning, lifelong learning, and critical thinking, understand the impact of technological changes, and adapt to new developments.

- Recognize the need for lifelong learning.
- Understand technological advances and their impact.

III. Duration of Study

Standard duration: 4 years

Study period: 3–6 years

IV. Main Discipline

Computer Science and Technology

V. Core Courses

C Programming, Discrete Mathematics, Computer Organization, Data Structures, Database Principles and Applications, Design and Analysis of Algorithms, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence.

VI. Main Practical Teaching Components

Military training, professional cognition practice, C programming course design, algorithm design course design, software engineering course design, intelligent application system development course design, professional internship, graduation design (thesis), second

classroom activities, etc.

VII. Graduation Requirements and Degree Awarding

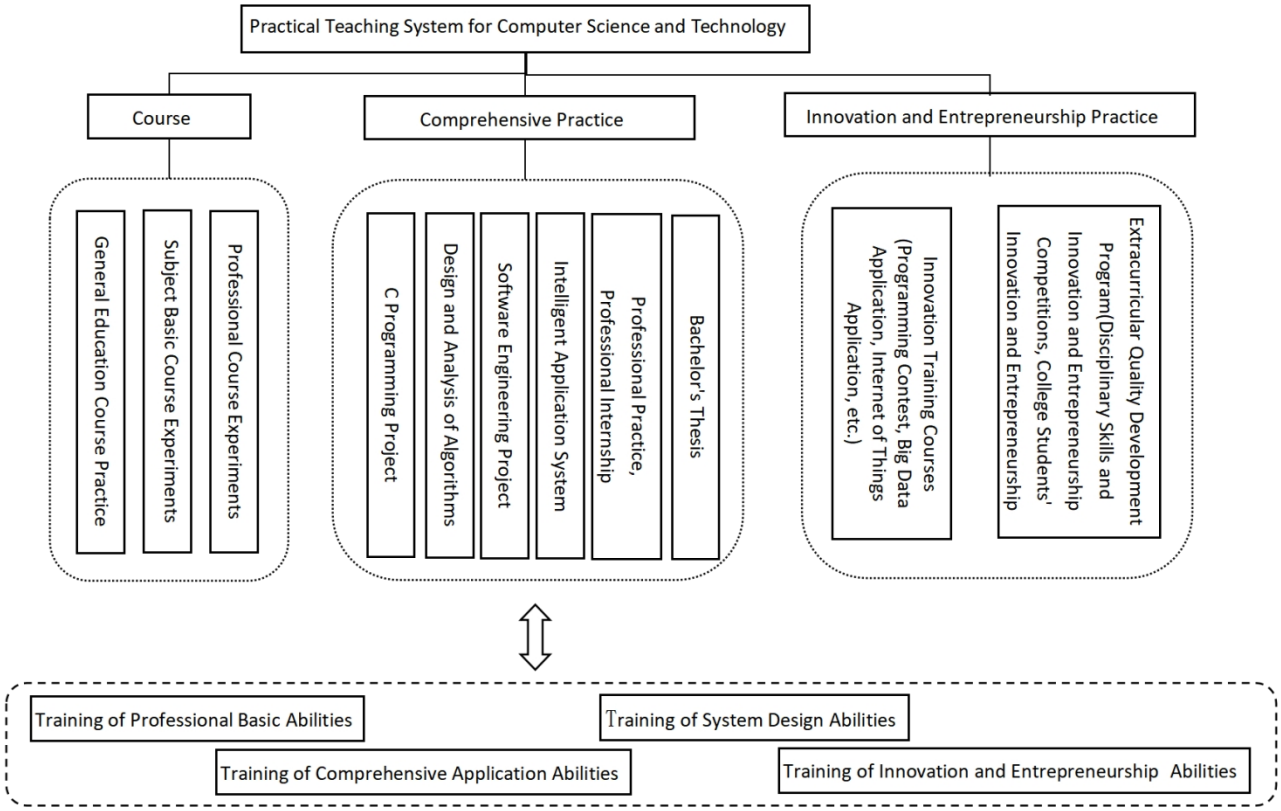
(1) Graduation Requirements

Within the prescribed study period, complete all courses specified in the training programme and pass all assessments. Obtain a minimum of 155 credits, including 137 credits of compulsory courses (including 24 credits of practical training) and at least 18 credits of elective courses.

(2) Degree Awarding

Students who meet the requirements for awarding a bachelor's degree shall be awarded the Bachelor of Engineering degree.

VIII. Flowchart of the Specialised Practical Training System



X. Curriculum of the Programme

Curriculum of Computer Science and Technology						
(Note: CP-Credit Point, S-Semester, L-Lecture, P-Practice)				Work Load		

Modules	Courses	Type	Chinese CP	ECTS CP	Contact Hours	Self-Study Hours	S	Remark
Mathematics and Physics	Advanced Mathematics A (1)	L	5	5	80	70	1	
	Advanced Mathematics A (2)	L	5	5	80	70	2	
	Linear Algebra A	L&P	3	3	48	42	1	
	College Physics B	L&P	4	4	64	56	2	
	Discrete Mathematics	L	3	3	48	42	2	
	Probability and Mathematical Statistics A	L&P	3	3	48	42	3	
Engineering Foundations	Python Programming	L&P	3	3	48	42	1	
	C Programming	L&P	4	4	64	56	2	
	Digital Logic Circuits	L&P	3	3	48	42	3	
	Data Structures	L&P	5	5	80	70	3	
	Database Principles and Applications	L&P	4	4	64	56	3	
	Computer Networks	L&P	4	4	64	56	4	
	Computer Organization and Architecture	L&P	5	5	80	70	4	
	Operating Systems	L	3	3	48	42	4	
	Introduction to Computer Science	L	1	1	16	14	1	
	Design and Analysis of Algorithms	L&P	3	3	48	42	4	
	Java Programming	L&P	3	3	48	42	4	
	Introduction to Artificial Intelligence	L&P	3	3	48	42	5	
	Principles of Compilers	L	2	2	32	28	5	
	Linux Operating System	L&P	2	2	32	28	5	
Engineering Application	Fundamentals of Data Mining	L&P	3	3	48	42	5	
	Software Engineering	L&P	3	3	48	42	6	
	Intelligent Application System Development	L&P	3	3	48	42	5	

	Advanced Java Programming	L&P	3	3	48	42	5	
	Mobile Application Development	L&P	3	3	48	42	4	
	Software Project Management	L&P	3	3	48	42	6	
	Human-Computer Interaction Technology	L&P	3	3	48	42	6	
	Machine Learning	L&P	3	3	48	42	6	
	Digital Image Processing	L&P	3	3	48	42	7	
	Data Visualization Techniques	L&P	3	3	48	42	7	
	Robot Technology and Applications	L&P	3	3	48	42	7	
	Professional English	L	2	2	32	28	6	
Electives	Fundamentals of Cloud Computing	L&P	2	2	32	28	6	15 ECTS credits are required
	Intelligent Healthcare Systems	L&P	2	2	32	28	7	
	Web Application Development	L&P	3	3	48	42	6	
	Innovative Training for Programming Contests	L&P	2	2	32	28	4	
	Cybersecurity	L&P	3	3	48	42	5	
	Scientific and Technical Writing	L	1	1	16	14	7	
	Fundamentals of Hadoop	L&P	2	2	32	28	6	
	Fundamentals of Spark	L&P	2	2	32	28	6	
	Big Data Development Technologies	L&P	2	2	32	28	6	
	Development of Applications Based on Large AI Models	L&P	2	2	32	28	7	
Foreign Language Courses	College English (1)	L	4	4	64	56	1	
	College English (2)	L	4	4	64	56	2	
General Courses	Ideological and Moral Cultivation and Legal Basis	L&P	3	3	48	42	1	
	Outline of Modern Chinese History	L	3	0	48	42	2	

	Fundamentals of Marxism	L	3	0	48	42	3	6 ECTS credits of other general elective courses are required
	Introduction to Mao Zedong Thought and the Theoretical System of Socialism with Chinese Characteristics	L&P	3	0	48	42	4	
	Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	L	3	0	48	42	5	
	Situation and Policy	L	2	0	64	56	1-8	
	College Physical Education (I)	L&P	1	2	36	24	1	
	College Physical Education (II)	L&P	1	2	36	24	2	
	College Physical Education (III)	L&P	1	2	36	24	3	
	College Physical Education (IV)	L&P	1	2	36	24	4	
	Military Theory	L	2	2	32	28	2	
	Career Planning	L	2	2	32	28	2	
	Fundamentals of Innovation for College Students	L	1	1	16	14	4	
	Career Guidance and Fundamentals of Entrepreneurship	L	1	1	16	14	5	
	Labor Education for College Students	L	1	1	16	14	1	
	Special Topics in National Security Education	L	1	1	16	14	1	
	Mental Health Education for College Students	L	2	4	60	60	1-7	
	Engineering Ethics	L	1	1	16	14	6	

	Military Training (Including Orientation Education)	P	2	0	60	60	1	
Practice Courses	Professional Practice	P	1	2	30	30	3-6	
	C Programming Project	P	1	2	30	30	2	
	Design and Analysis of Algorithms Project	P	2	4	60	60	5	
	Software Engineering Project	P	2	4	60	60	6	
	Intelligent Application System Development Project	P	1	2	30	30	6	
	Professional Internship	P	3	6	90	90	7	
	Extracurricular Quality Development Program	P	4	8	120	120	1-8	
Bachelor's Thesis	Bachelor's Thesis	P	12	28	240	600	8	
Total ECTS: 212, ECTS/PER SEMISTER:26.5								

师资队伍	师资情况											
			职称结构				学历结构			参与国际交流或境外学习经历	科技副总	人才称号
	信息工程学院师资结构	类别	正高	副高	讲师	助教及以下	博士	博士在读及硕士	硕士			
		数量	5	18	25	4	26	1	17	7	2	6
			职称结构				学历结构					
	计算机科学与技术专业师资结构	类别	正高	副高	讲师	助教及以下	博士	博士在读及硕士	硕士	参与国际交流或境外学习经历	科技副总	人才称号
		数量	2	4	10	0	8	0	8	2	1	3

教师科研情况

类别	国家一流课程	省一流课程	市厅级项目	横向项目（到账经费）	授权发明/实用新型专利	发表期刊论文	科研获奖
数量	6	45	37	127（1805.099615万元）	89（11/78）	329	17

课程建设

类别	国家一流课程	省级一流课程	省级一平台两系统	省级在线开放课程
数量	1	5	1	2

教师教育教学成果

类别	教学竞赛	指导学科竞赛	教学成果奖	教学研究项目	教学改革论文	教材
数量	70	412	6	71	18	5

资源建设

计算机科学与专业图书资料情况

中文藏书	外文藏书	订阅国内专业期刊（种）	订阅国外专业	中文数据库数（个）	外文数据库数（个）	电子期刊
134452	5176	829	2	36	9	851

学科平台、实验室资源

类别	科研平台			教学实验室				教学实验基地	
	省级平台	市级平台	合计	专业实验室	教学仪器设备	在校生数	生均教学仪器设备经费（万）	原有实践基地	新建实践基地
数量	1	4	5	13	1251	13500	1.97	28	2

专业建设经费

项目	课程建设费（万）	教师发展与教学团队建设（万）	教材资源开发（万）	实验实训条件建设（万）	学生创新创业训练（万）	国内外教学交流合作（万）	教育教学研究与改革（万）	合计（万）
金额	14.1	80	60	980.2	123.8	70	167.1	1495.2

人才培养方案情况

类别	中国学分（CP）	欧洲学分（ECTS）	实践学分（ECTS）	与企业合作实验学分（ECTS）	与企业共同开展实践课程(个)	与企业共建专业课程(个)	与企业共带毕业设计(个
金额	155	212	55	0	0	0	6

学生学位授予、就业、考研

2023 年	学生人数	42	2024 年	学生人数	47	2025 年	学生人数	88
	毕业获学位人数	36		毕业获学位人数	47		毕业获学位人数	88
	学位授予率	86%		学位授予率	98%		学位授予率	100%
	考研录取人数	8		考研录取人数	17		考研录取人数	16
	考研率	19%		考研率	37%		考研率	18%
	就业人数	36		就业人数	47		就业人数	88
	就业率	86%		就业率	100%		就业率	100%

学生入学及在校情况统计表

入学年份	总招生人数	按时毕业人数	推迟毕业人数	当年在校人数
2016	48			126
2017	53			179
2018	39			218
2019	42	78		182
2020	47	44	3	188
2021	88	52	4	216
2022	156	38	1	333
2023	199	36	0	490
2024	204	45	1	650
2025	201	90	1	764

学生入学录取率统计表

入学年份	申请人数	招生名额	录取学生数	录取率（%）
2021		200	200	100
2022		204	204	100
2023		204	204	100
2024		160	160	100
2025		88	88	100

学生科创实践活动统计

类别	学科竞赛获奖（人次）	大学生创新创业项目（人次）	学生参与老师科研项目	学生专利	参加社会实实践活动
人数	325	168	13	32	每学年开展40 次

学生流动统计表

入学年份	转入学生	转出学生
2021	0	5
2022	4	9
2023	4	12
2024	3	4
2025	5	5

交流项目概览

交流项目名称	简述	近五年参与人数
澳门暑期文化交流项目	在澳门开展的短期暑期文化交流项目	1
与澳门城市大学人工智能交流项目	聚焦人工智能的短期学术交流项目	2
与澳大利亚麦考瑞大学人工智能交流项目	与麦考瑞大学开展的人工智能领域短期学术交流项目	6
与澳大利亚麦考瑞大学人工智能课程项目	在麦考瑞大学开展的人工智能课程制国际项目	2